



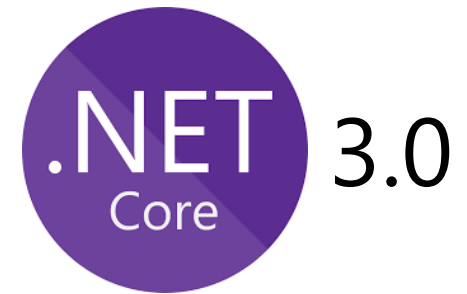
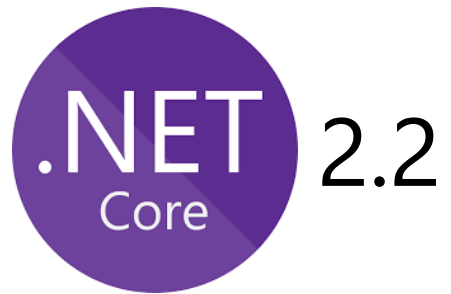
TRAILHEAD
TECHNOLOGY PARTNERS

When Bad Libraries Happen to Good Code

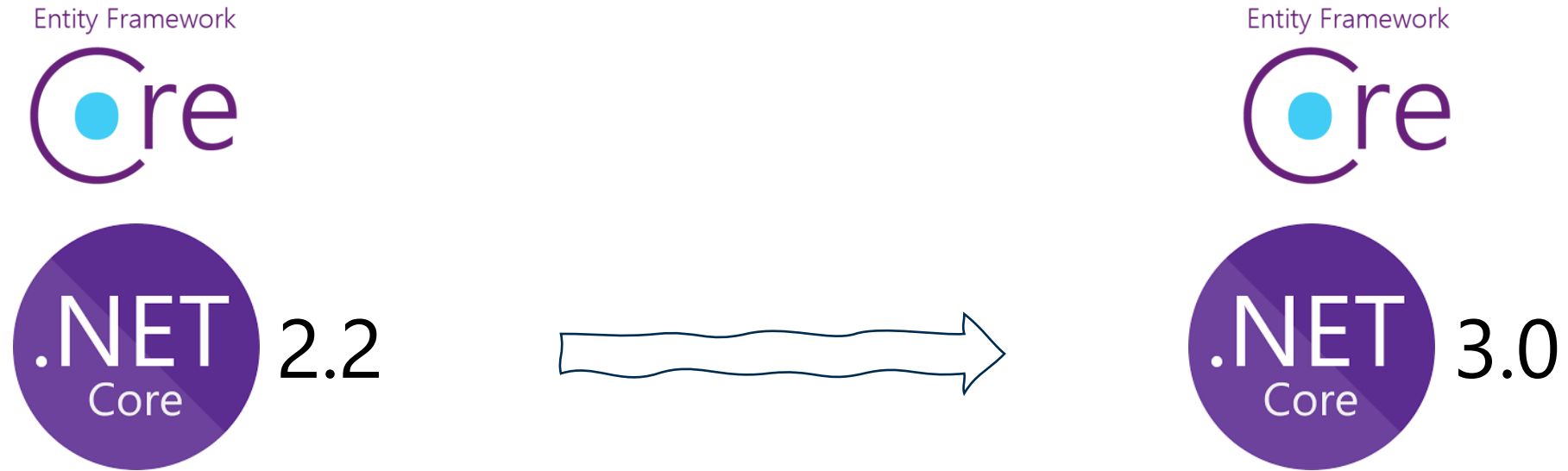


Jonathan "J." Tower

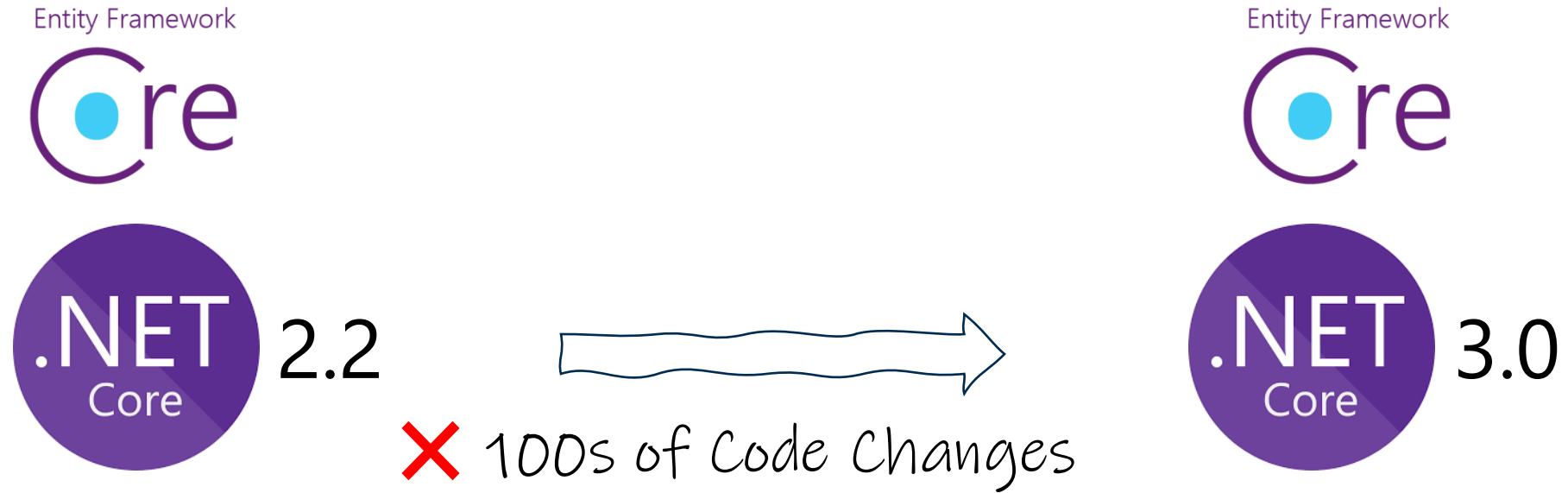
Client A



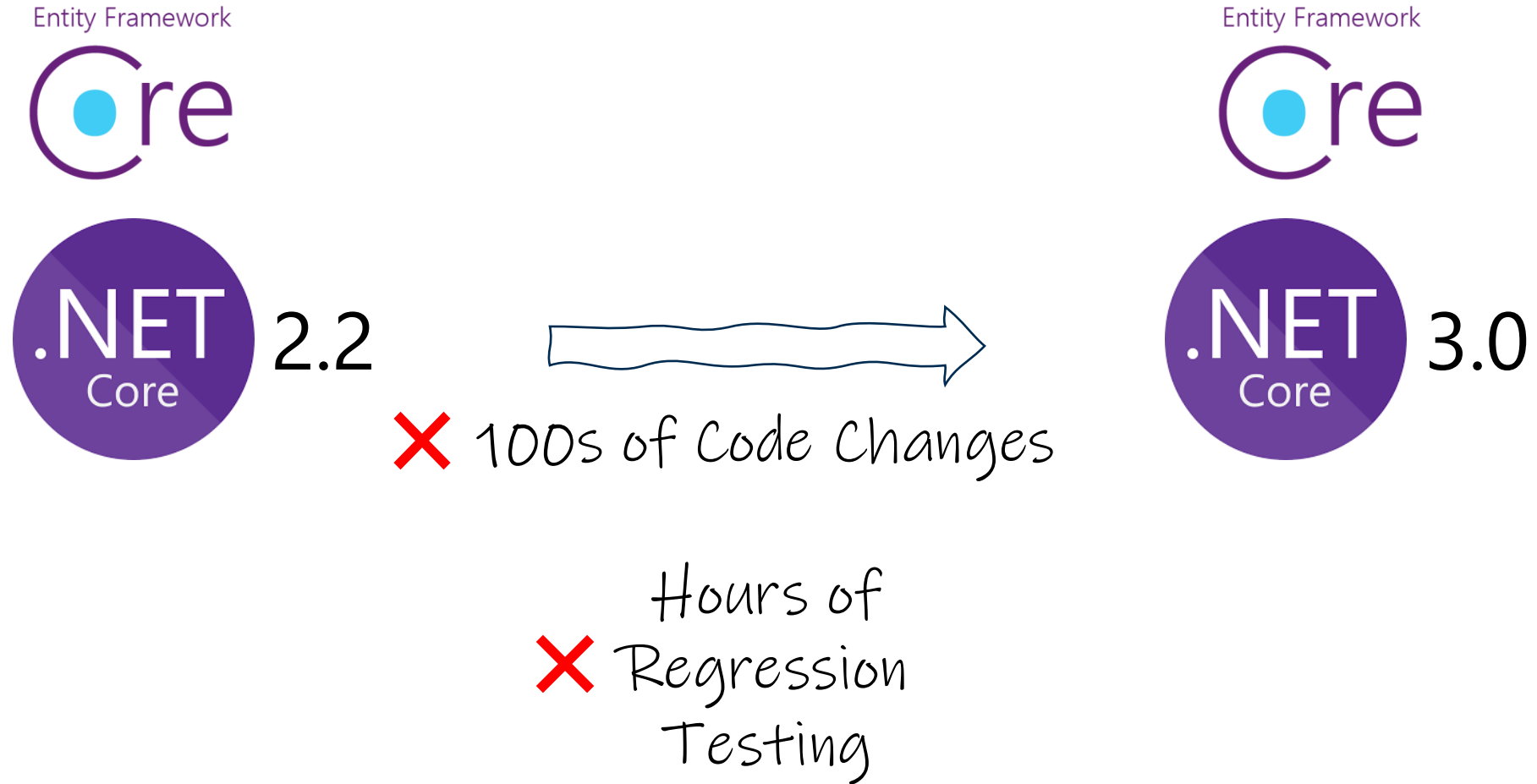
Client A



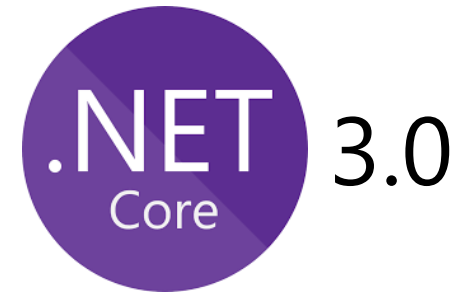
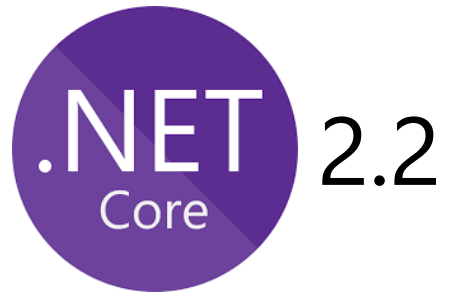
Client A



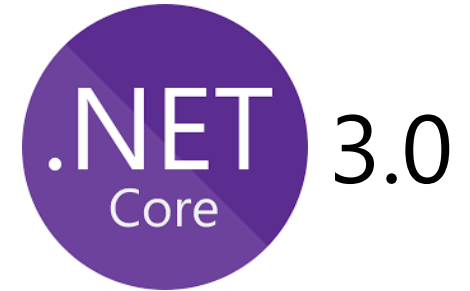
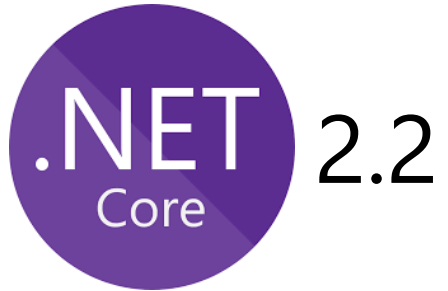
Client A



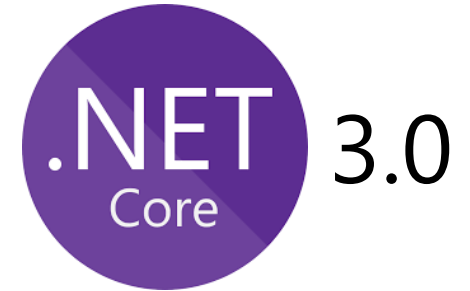
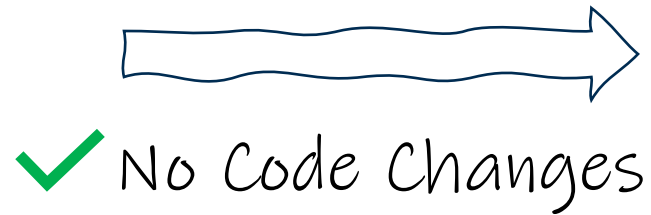
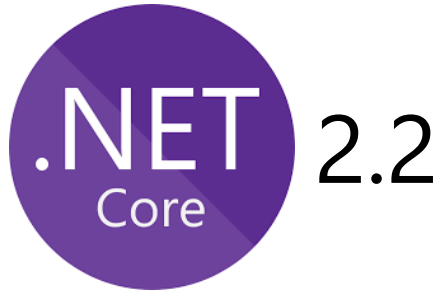
Client B



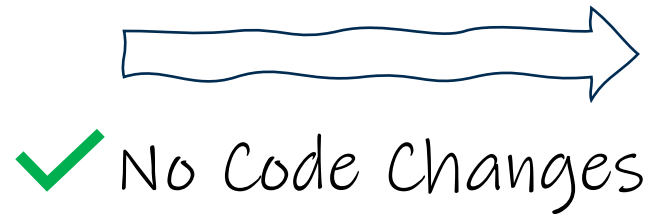
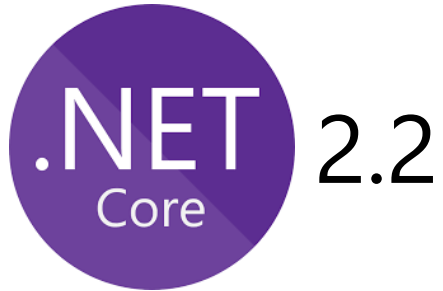
Client B



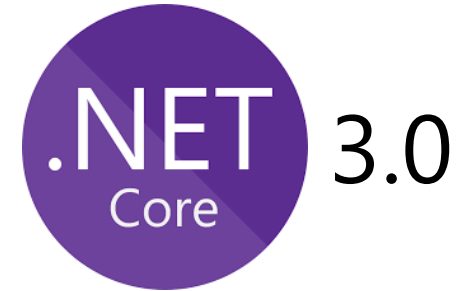
Client B



Client B



✓ Everything just worked





Your **Choice** of
Library **Matters!**



TRAILHEAD
TECHNOLOGY PARTNERS

How to **choose** and **manage** third-party libraries through a simple framework that makes **regret less likely** later

Jonathan "J." Tower

Principal Consultant & Partner



🏆 Microsoft MVP in .NET

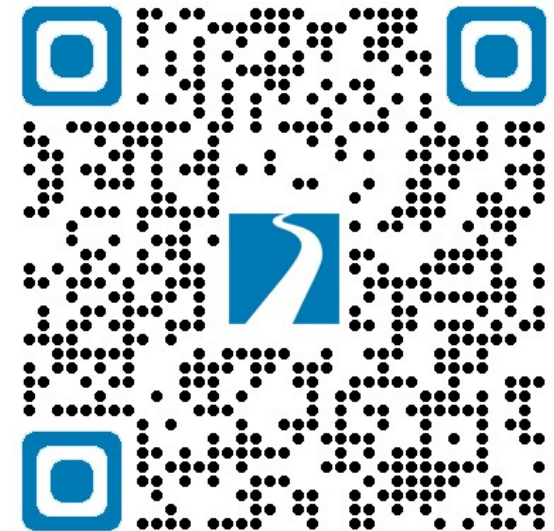
✉ jtower@trailheadtechnology.com

🌐 trailheadtechnology.com/blog

🐦 jtowermi

in jtower

EXPERT CONSULTATION



bit.ly/th-offer

github.com/trailheadtechnology/3rd-party-lib

"Third-Party"

What Does It Mean?

What Is Third-Party?

1. Relating to a person or group **besides the two primarily** involved in a situation:

- Not from you
- Not from your main Framework (.NET, Microsoft, etc)

2. Any **optional package** you choose to add that isn't required by the base framework/runtime.

What Is Third-Party

1. Relating to a person or group **besides the two primarily** involved in a situation:
 - Not from you
 - Not from your main Framework (.NET, Microsoft, etc)
2. Any **optional package** you choose to add that isn't required by the base framework/runtime.

The Basics

The Three Areas You're Probably Already Thinking About



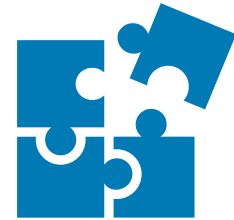
The Basics



Health of
the Project



Safety &
Reliability



Fit for
Purpose

Health of the Project

Is it alive and sustainable?

Popularity

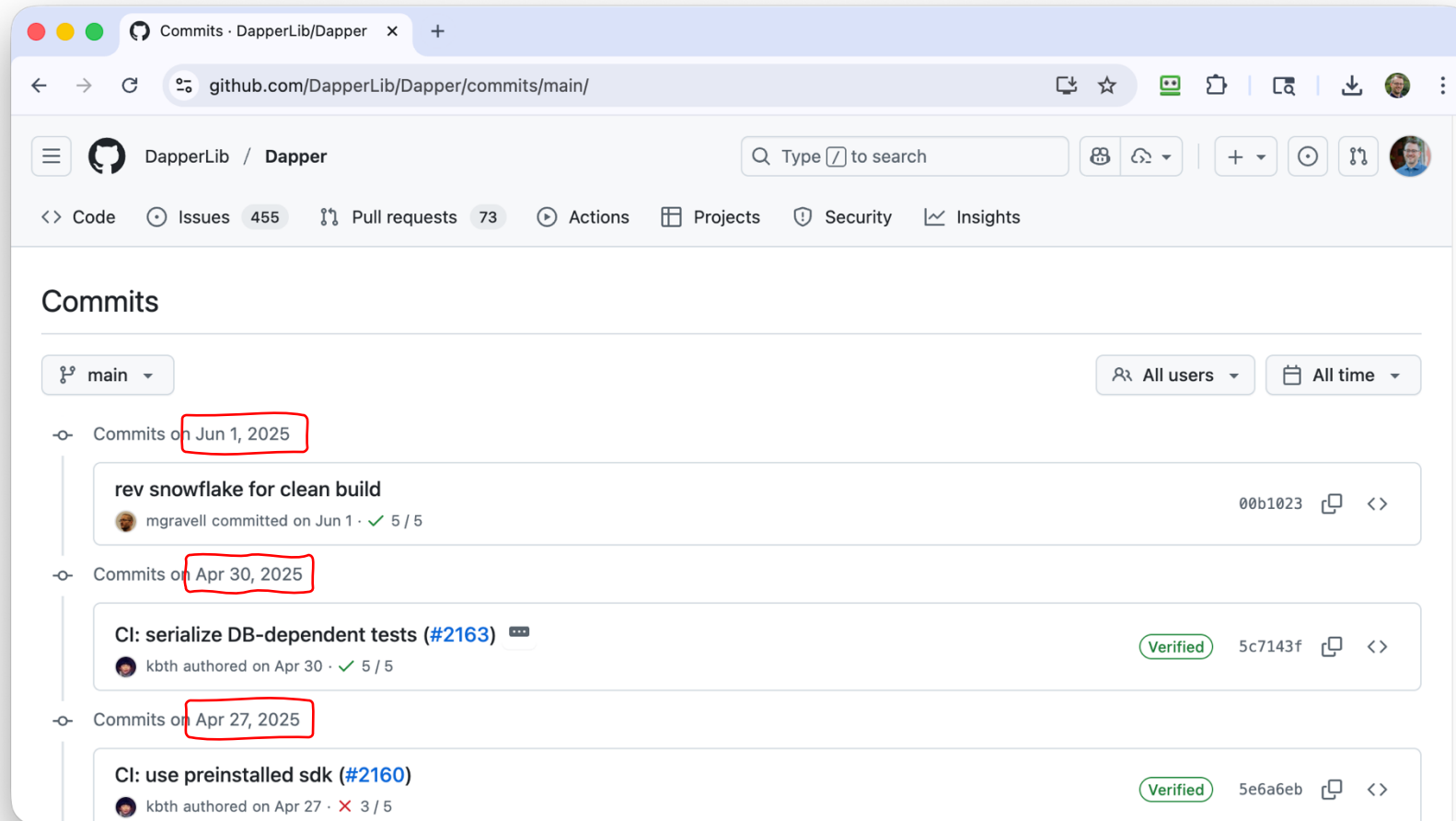
The screenshot shows the GitHub repository page for **DapperLib/Dapper**. The repository is public and has the following metrics:

- Stars:** 18.1k
- Forks:** 3.7k
- Watching:** 919

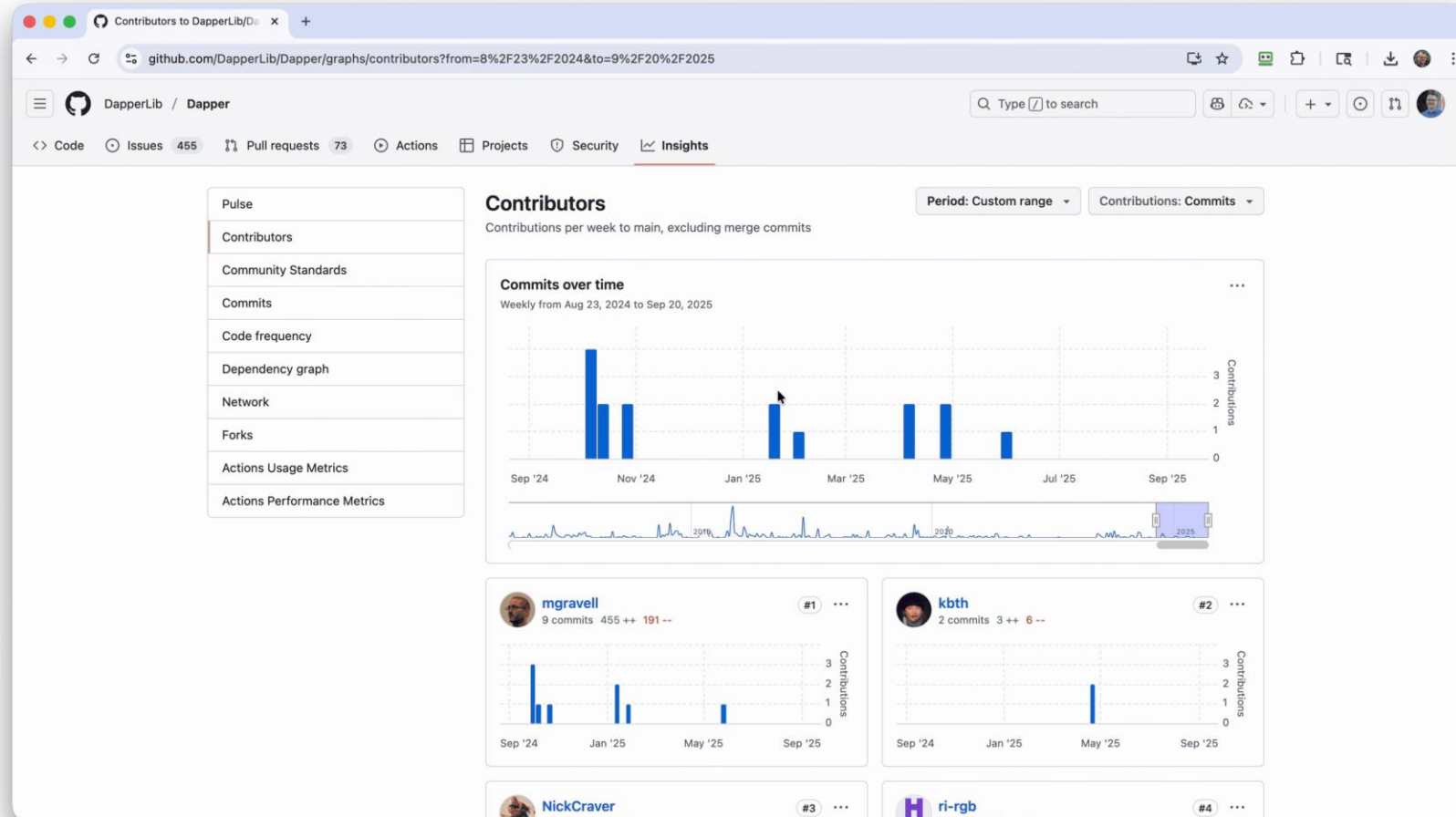
A red box highlights the 'Watch', 'Fork', and 'Star' buttons and their respective counts. The repository is described as 'Dapper - a simple object mapper for .Net' and includes a link to www.learnmapper.com/. The 'About' section lists features like Readme, View license, Activity, Custom properties, 18.1k stars, 919 watching, 3.7k forks, and a report repository. The 'Releases' section shows the latest version, 2.1.66, released on Feb 6.

File/Folder	Description	Last Updated
.github	CI: serialize DB-dependent tests (#2163)	5 months ago
Dapper.EntityFramework.StrongName	attempt NRT makeover (#1928)	2 years ago
Dapper.EntityFramework	attempt NRT makeover (#1928)	2 years ago
Dapper.ProviderTools	normalize async API surface over all TFMs (#2144)	7 months ago
Dapper.Rainbow	Issue #2044 (#2045)	last year
Dapper.SqlBuilder	Fix and clarify OrWhere caveats in SqlBuilder docs (#2149)	5 months ago
Dapper.StrongName	normalize async API surface over all TFMs (#2144)	7 months ago
Dapper	OutputExpression wasn't working for methods that use Q...	5 months ago
benchmarks	normalize async API surface over all TFMs (#2144)	7 months ago
docs	change dapper-plus citation (#2083)	last year
signatures/version1	Creating file for storing CLA Signatures	last year

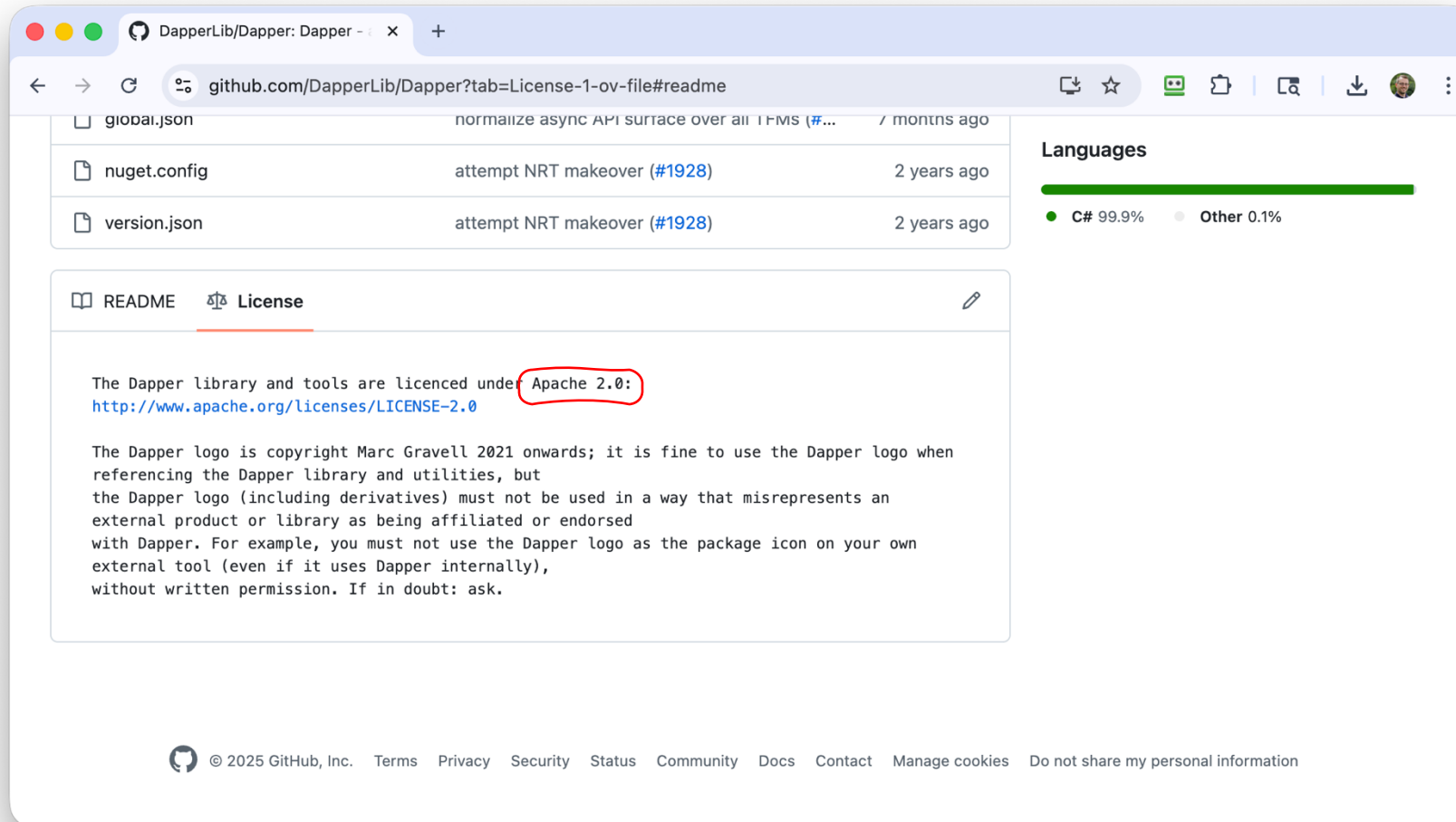
Active Maintenance



Community & Ecosystem



Long-Term Support & Exit Strategy



The screenshot shows the GitHub repository page for DapperLib/Dapper. The browser address bar displays `github.com/DapperLib/Dapper?tab=License-1-ov-file#readme`. The repository files list includes `global.json`, `nuget.config`, and `version.json`. The `License` tab is selected, showing the Apache 2.0 license, which is circled in red. The license text states: "The Dapper library and tools are licenced under Apache 2.0: <http://www.apache.org/licenses/LICENSE-2.0>". Below this, there is a paragraph about the Dapper logo copyright. The right sidebar shows the "Languages" section with a bar chart indicating that 99.9% of the code is in C# and 0.1% is in Other. The footer of the page includes the GitHub logo, copyright information for 2025, and links to Terms, Privacy, Security, Status, Community, Docs, Contact, Manage cookies, and Do not share my personal information.

File	Description	Last Updated
<code>global.json</code>	normalize async API surface over all TFMS (#...	7 months ago
<code>nuget.config</code>	attempt NRT makeover (#1928)	2 years ago
<code>version.json</code>	attempt NRT makeover (#1928)	2 years ago

Languages

● C# 99.9% ● Other 0.1%

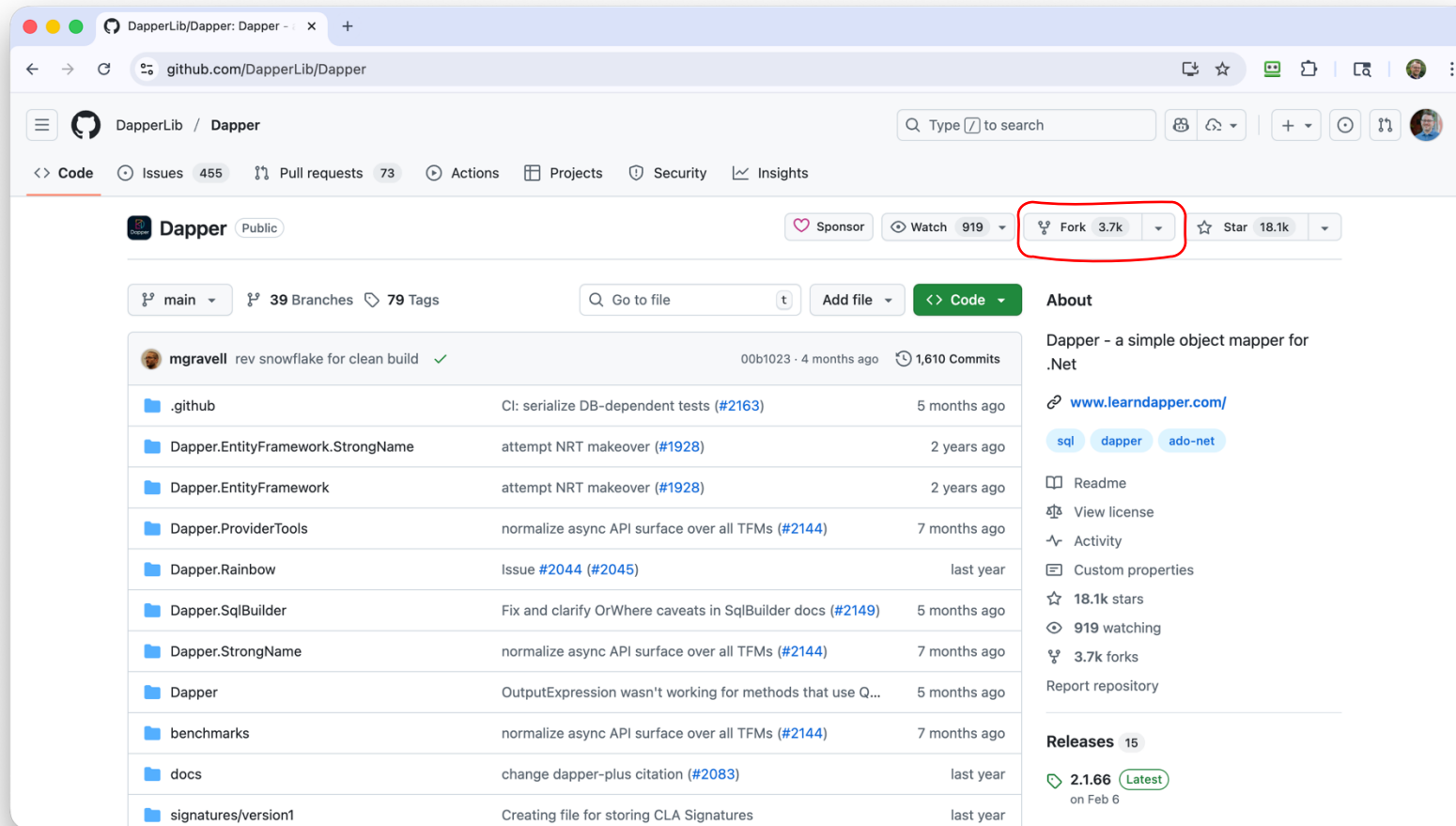
License

The Dapper library and tools are licenced under Apache 2.0: <http://www.apache.org/licenses/LICENSE-2.0>

The Dapper logo is copyright Marc Gravell 2021 onwards; it is fine to use the Dapper logo when referencing the Dapper library and utilities, but the Dapper logo (including derivatives) must not be used in a way that misrepresents an external product or library as being affiliated or endorsed with Dapper. For example, you must not use the Dapper logo as the package icon on your own external tool (even if it uses Dapper internally), without written permission. If in doubt: ask.

© 2025 GitHub, Inc. [Terms](#) [Privacy](#) [Security](#) [Status](#) [Community](#) [Docs](#) [Contact](#) [Manage cookies](#) [Do not share my personal information](#)

Long-Term Support & Exit Strategy



The screenshot shows the GitHub repository page for DapperLib/Dapper. The repository is public and has 18.1k stars, 919 watchers, and 3.7k forks. The 'Fork' button is highlighted with a red rectangle. The repository is maintained by mgravell and has 1,610 commits. The file list on the left includes .github, Dapper.EntityFramework.StrongName, Dapper.EntityFramework, Dapper.ProviderTools, Dapper.Rainbow, Dapper.SqlBuilder, Dapper.StrongName, Dapper, benchmarks, docs, and signatures/version1. The right sidebar shows the repository's description, links to the README, license, and activity, as well as the latest release (2.1.66) dated Feb 6.

Repository: DapperLib / Dapper

Stats: 18.1k Stars, 919 Watchers, 3.7k Forks

Files:

- .github: CI: serialize DB-dependent tests (#2163) 5 months ago
- Dapper.EntityFramework.StrongName: attempt NRT makeover (#1928) 2 years ago
- Dapper.EntityFramework: attempt NRT makeover (#1928) 2 years ago
- Dapper.ProviderTools: normalize async API surface over all TFMs (#2144) 7 months ago
- Dapper.Rainbow: Issue #2044 (#2045) last year
- Dapper.SqlBuilder: Fix and clarify OrWhere caveats in SqlBuilder docs (#2149) 5 months ago
- Dapper.StrongName: normalize async API surface over all TFMs (#2144) 7 months ago
- Dapper: OutputExpression wasn't working for methods that use Q... 5 months ago
- benchmarks: normalize async API surface over all TFMs (#2144) 7 months ago
- docs: change dapper-plus citation (#2083) last year
- signatures/version1: Creating file for storing CLA Signatures last year

About: Dapper - a simple object mapper for .Net

Links: www.learnmapper.com/, sql, dapper, ado-net

Releases: 15

Latest Release: 2.1.66 (Latest) on Feb 6

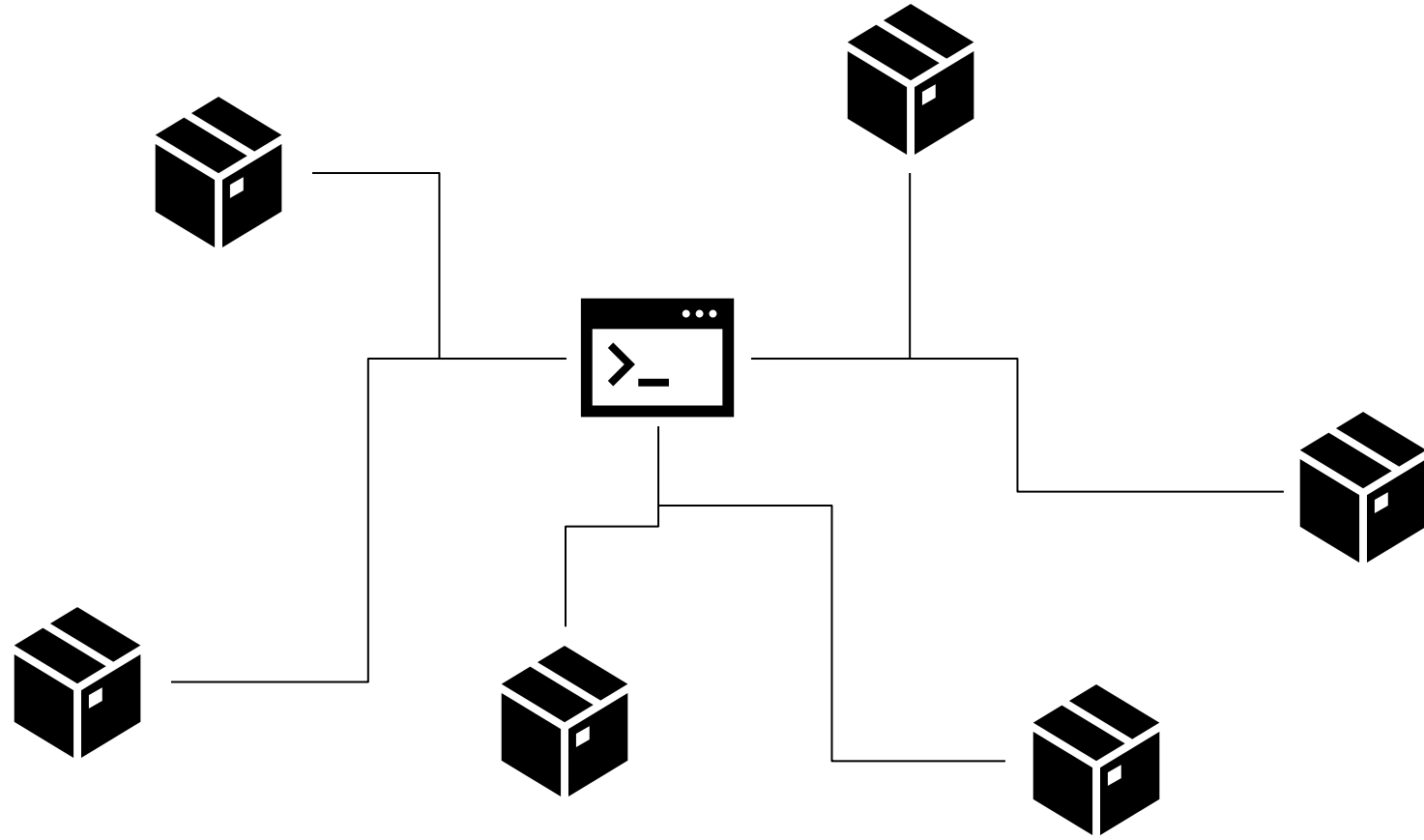
Safety & Reliability

Will it burn us later?

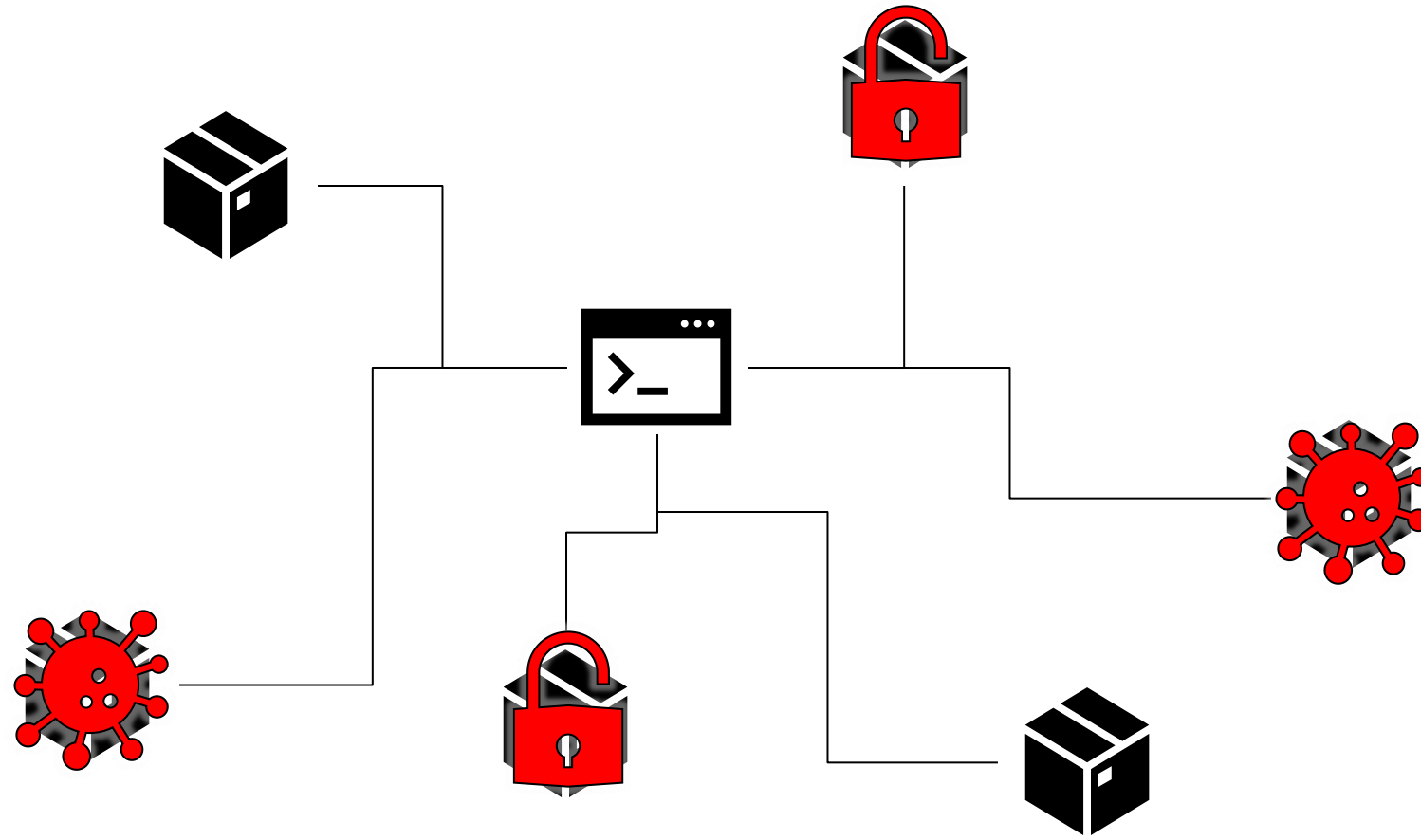
License

Category	Licenses	Commercial Use Notes
 Safe / Permissive	MIT, Apache 2.0 BSD (2/3-Clause) ISC	Free to use in commercial apps. No obligation to open-source your code. Apache adds patent protection.
 Caution	LGPL MPL	Allowed in commercial apps, but rules on linking/modification. Must share changes to the library itself.
 Risky / Copyleft	GPL AGPL	If you distribute software using these, you may need to open-source <i>your entire app</i> . Usually avoided in commercial projects.

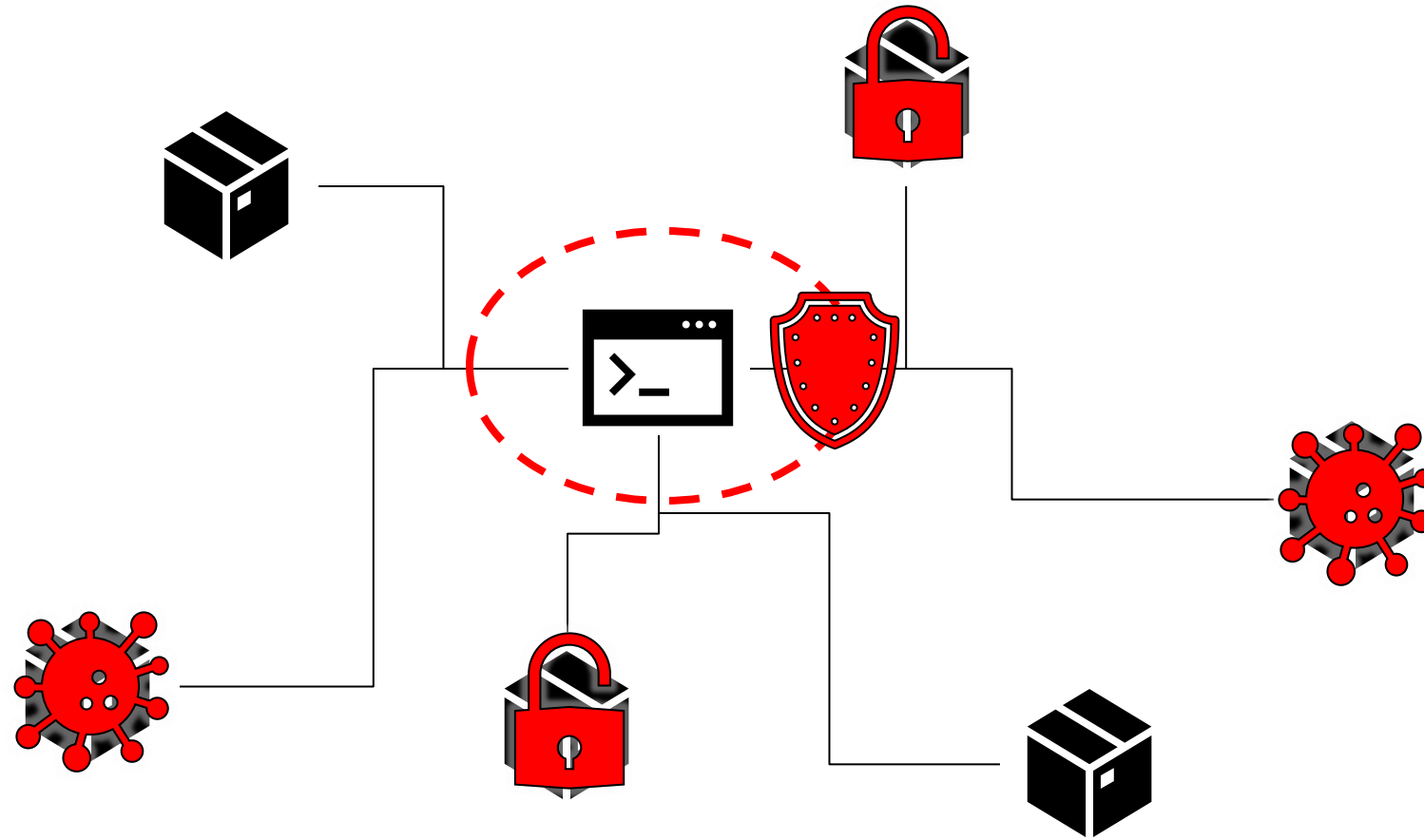
Security



Security



Security



Security

The screenshot shows the NuGet Package Manager interface. On the left, a list of packages is displayed, including Newtonsoft.Json, Newtonsoft.Json.Bson, Newtonsoft.Json.Schema, Microsoft.AspNetCore.Mvc.Newtonsoft.Json, Swashbuckle.AspNetCore.Newtonsoft, GraphQL.Newtonsoft.Json, RestSharp.Newtonsoft.Json, App.Metrics.Formatters.Json, NServiceBus.Newtonsoft.Json, Refit.Newtonsoft.Json, and RestSharp.Serializers.Newtonsoft.Json. On the right, the details for the selected package, Newtonsoft.Json, are shown. A red circle highlights the 'Vulnerabilities detected (1)' section, which indicates that the current version (12.0.3) has at least one vulnerability with high severity. The details pane also shows the version dropdown set to 12.0.3, the 'Install' button, and various metadata fields like Author(s), License, Date published, Project URL, Report Abuse, Tags, and Dependencies.

NuGet Package Manager: Sample

Package source: nuget.org

Newtonsoft.Json by James Newton-King, 2.98 downloads

Json.NET is a popular high-performance JSON framework for .NET

Version: 12.0.3 Install

Vulnerabilities detected (1)

This package version has at least one vulnerability with high severity. It may lead to specific problems in your project. Try updating the package version.

Details

Advisory (high severity): <https://github.com/advisories/GHSA-5crp-9r3c-p9vr>

Options

Description

Json.NET is a popular high-performance JSON framework for .NET

Version: 12.0.3

Author(s): James Newton-King

License: MIT

Date published: Friday, November 8, 2019 (11/8/2019)

Project URL: <https://www.newtonsoft.com/json>

Report Abuse: <https://www.nuget.org/packages/Newtonsoft.Json/12.0.3/ReportAbuse>

Tags: json

Dependencies

- .NETFramework,Version=v2.0
 - No dependencies
- .NETFramework,Version=v3.5
 - No dependencies
- .NETFramework,Version=v4.0
 - No dependencies
- .NETFramework,Version=v4.5
 - No dependencies
- .NETPortable,Version=v0.0,Profile=Profile259
 - No dependencies
- .NETPortable,Version=v0.0,Profile=Profile328
 - No dependencies
- .NETStandard,Version=v1.0
 - Microsoft.CSharp (>= 4.3.0)
 - NETStandard.Library (>= 1.6.1)
 - System.ComponentModel.TypeConverter (>= 4.3.0)

Security

`dotnet list package --vulnerable`

```
eShopCoreMVC — zsh — 117x30
jonathantower@MacBookPro eShopCoreMVC % dotnet list package --vulnerable

The following sources were used:
  https://api.nuget.org/v3/index.json

Project `eShopCoreMVC` has the following vulnerable packages
[net6.0]:
Top-level Package      Requested  Resolved  Severity  Advisory URL
> Yarp.ReverseProxy    2.0.0     2.0.0     High      https://github.com/advisories/GHSA-jrjw-qgr2-wfcg

jonathantower@MacBookPro eShopCoreMVC %
```

Security

`dotnet list package --vulnerable`

```
jonathantower@MacBookPro eShopCoreMVC % dotnet list package --vulnerable

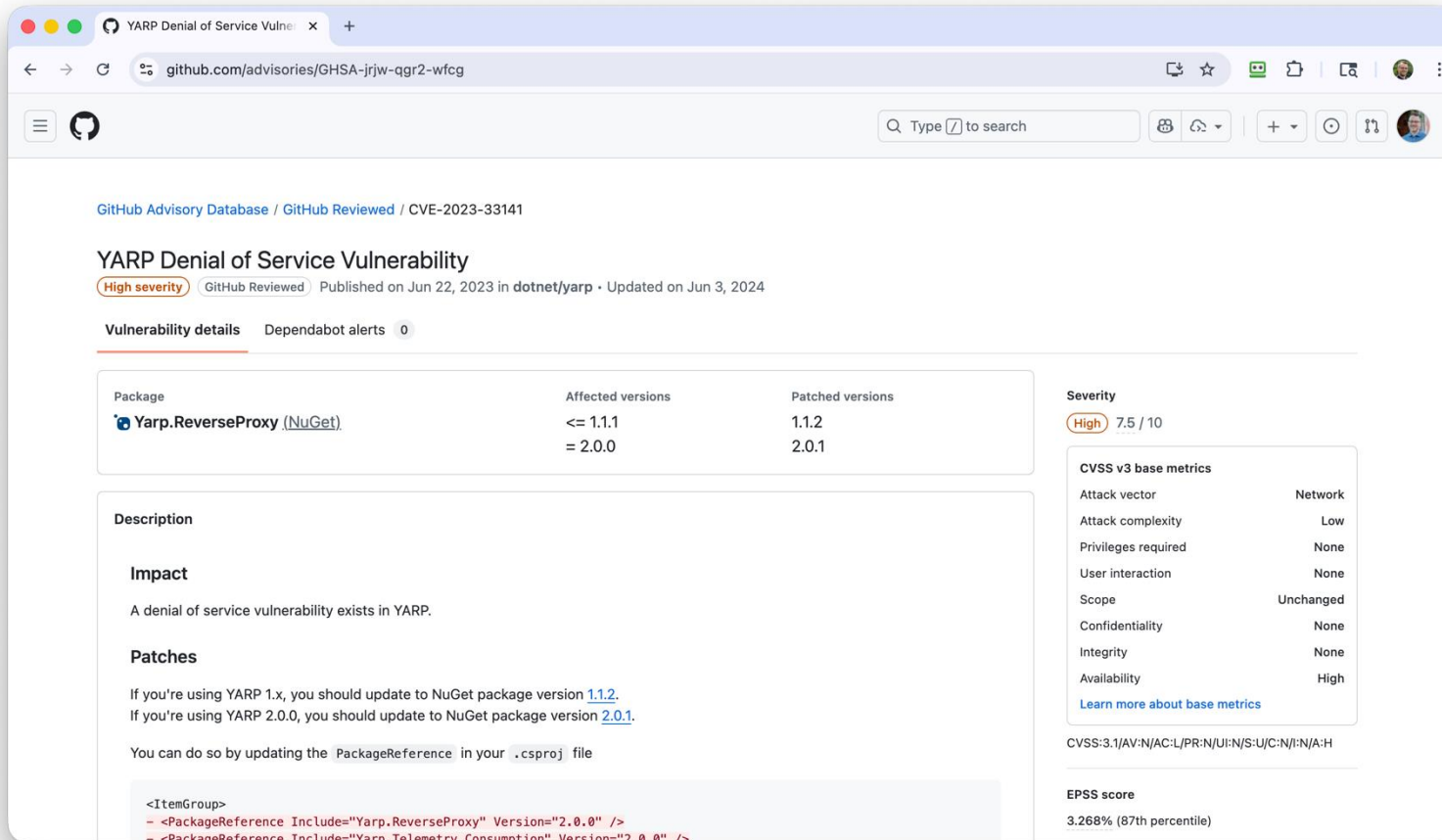
The following sources were used:
https://api.nuget.org/v3/index.json

Project `eShopCoreMVC` has the following vulnerable packages
[net6.0]:
Top-level Package      Requested  Resolved  Severity  Advisory URL
> Yarp.ReverseProxy    2.0.0     2.0.0     High      https://github.com/advisories/GHSA-jrjw-qgr2-wfcg

jonathantower@MacBookPro eShopCoreMVC %
```

Safety & Reliability: Security

dotnet list package --vulnerable



The screenshot shows a web browser displaying a GitHub Advisory Database entry for a Denial of Service (DoS) vulnerability in YARP. The page title is "YARP Denial of Service Vulnerability" with a "High severity" label. It was published on June 22, 2023, in the dotnet/yarp repository and updated on June 3, 2024. The vulnerability details table shows that the affected versions are <= 1.1.1 and = 2.0.0, while patched versions are 1.1.2 and 2.0.1. The severity is rated as High (7.5/10). The CVSS v3 base metrics are listed as: Attack vector (Network), Attack complexity (Low), Privileges required (None), User interaction (None), Scope (Unchanged), Confidentiality (None), Integrity (None), and Availability (High). The EPSS score is 3.268% (87th percentile). The description states that a denial of service vulnerability exists in YARP. The patches section provides instructions for updating YARP 1.x to NuGet package version 1.1.2 and YARP 2.0.0 to NuGet package version 2.0.1. A code snippet shows how to update the PackageReference in a .csproj file.

GitHub Advisory Database / GitHub Reviewed / CVE-2023-33141

YARP Denial of Service Vulnerability

High severity GitHub Reviewed Published on Jun 22, 2023 in dotnet/yarp • Updated on Jun 3, 2024

Vulnerability details Dependabot alerts 0

Package	Affected versions	Patched versions
 Yarp.ReverseProxy (NuGet)	<= 1.1.1 = 2.0.0	1.1.2 2.0.1

Description

Impact

A denial of service vulnerability exists in YARP.

Patches

If you're using YARP 1.x, you should update to NuGet package version [1.1.2](#).
If you're using YARP 2.0.0, you should update to NuGet package version [2.0.1](#).

You can do so by updating the `PackageReference` in your `.csproj` file

```
<ItemGroup>
  - <PackageReference Include="Yarp.ReverseProxy" Version="2.0.0" />
  - <PackageReference Include="Yarp.Telemetry.Consumption" Version="2.0.0" />
```

Severity

High 7.5 / 10

CVSS v3 base metrics

Metric	Value
Attack vector	Network
Attack complexity	Low
Privileges required	None
User interaction	None
Scope	Unchanged
Confidentiality	None
Integrity	None
Availability	High

[Learn more about base metrics](#)

CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

EPSS score

3.268% (87th percentile)

Safety & Reliability: Security

dotnet list package --vulnerable

The screenshot shows a web browser displaying a GitHub Advisory Database entry for a Denial of Service (DoS) vulnerability in YARP. The browser's address bar shows the URL `github.com/advisories/GHSA-jrjw-qgr2-wfcg`. The page title is "YARP Denial of Service Vulnerability" with a "High severity" label. It was published on June 22, 2023, in the `dotnet/yarp` repository and updated on June 3, 2024. The "Vulnerability details" section shows the package `Yarp.ReverseProxy` (NuGet) with affected versions `<= 1.1.1` and `= 2.0.0`. The patched versions are `1.1.2` and `2.0.1`, which are highlighted with a red circle. The "Description" section includes an "Impact" stating a denial of service vulnerability exists in YARP, and "Patches" advising users to update to NuGet package version 1.1.2 or 2.0.1. The "Severity" section shows a CVSS v3 base metrics score of 7.5/10 (High) and an EPSS score of 3.268% (87th percentile). The CVSS vector is `CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H`. The "Description" section also includes a code snippet for updating the `PackageReference` in a `.csproj` file.

GitHub Advisory Database / GitHub Reviewed / CVE-2023-33141

YARP Denial of Service Vulnerability

High severity GitHub Reviewed Published on Jun 22, 2023 in dotnet/yarp • Updated on Jun 3, 2024

Vulnerability details Dependabot alerts 0

Package	Affected versions	Patched versions
Yarp.ReverseProxy (NuGet)	<code><= 1.1.1</code> <code>= 2.0.0</code>	1.1.2 2.0.1

Description

Impact

A denial of service vulnerability exists in YARP.

Patches

If you're using YARP 1.x, you should update to NuGet package version [1.1.2](#).
If you're using YARP 2.0.0, you should update to NuGet package version [2.0.1](#).

You can do so by updating the `PackageReference` in your `.csproj` file

```
<ItemGroup>
  - <PackageReference Include="Yarp.ReverseProxy" Version="2.0.0" />
  - <PackageReference Include="Yarp.Telemetry.Consumption" Version="2.0.0" />
  + <PackageReference Include="Yarp.ReverseProxy" Version="1.1.2" />
  + <PackageReference Include="Yarp.Telemetry.Consumption" Version="2.0.1" />
</ItemGroup>
```

Severity

High 7.5 / 10

CVSS v3 base metrics

Attack vector	Network
Attack complexity	Low
Privileges required	None
User interaction	None
Scope	Unchanged
Confidentiality	None
Integrity	None
Availability	High

[Learn more about base metrics](#)

CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

EPSS score

3.268% (87th percentile)

Security

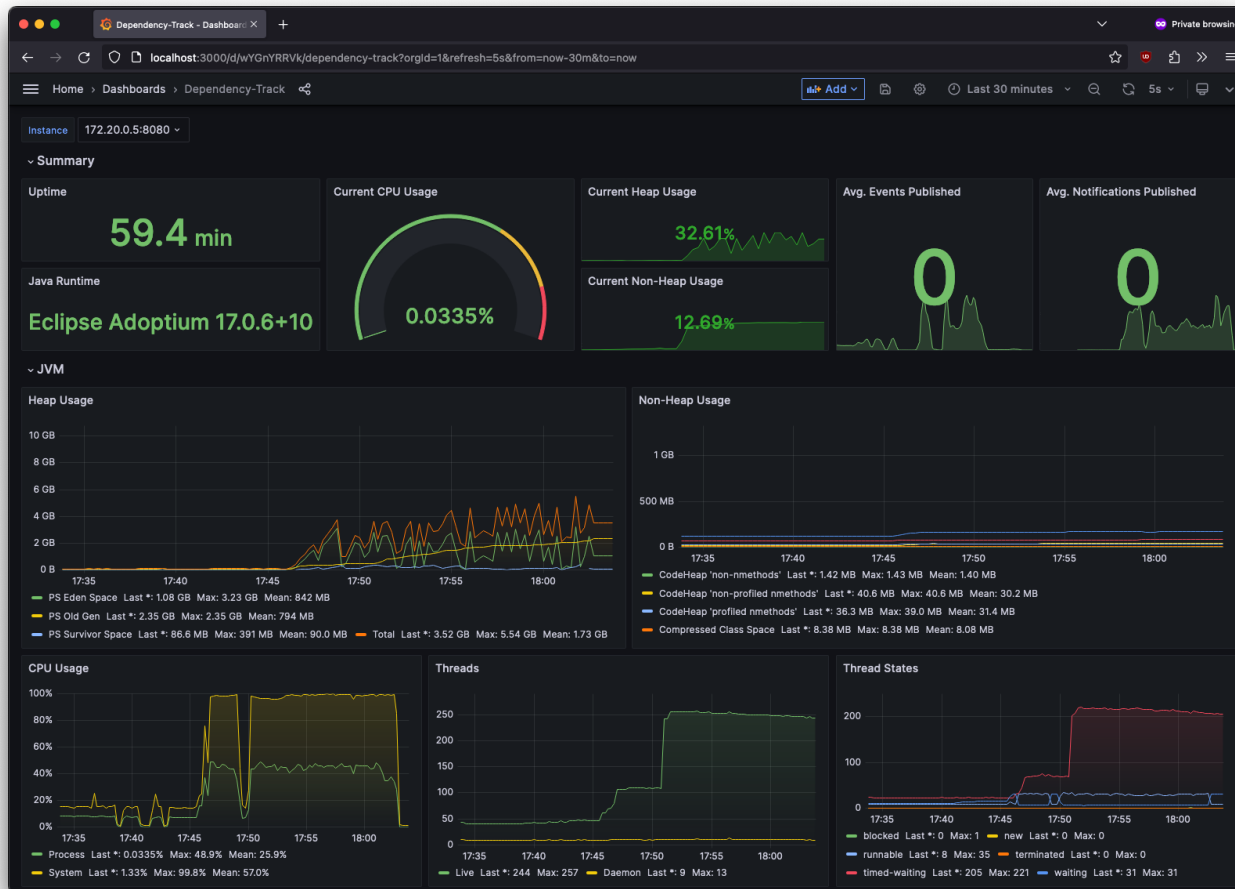
```
# npm audit report

ansi-html *
Severity: high
Uncontrolled Resource Consumption in ansi-html - https://github.com/advisories/GHSA-whgm-jr23-g3j9
fix available via `npm audit fix --force`
Will install @vue/cli-plugin-babel@3.12.1, which is a breaking change
node_modules/ansi-html
  webpack-dev-server 2.0.0-beta - 4.1.0
    Depends on vulnerable versions of ansi-html
    Depends on vulnerable versions of chokidar
    Depends on vulnerable versions of http-proxy-middleware
    Depends on vulnerable versions of yargs
  node_modules/webpack-dev-server
    @vue/cli-service <=5.0.0-beta.3
      Depends on vulnerable versions of @vue/cli-plugin-router
      Depends on vulnerable versions of @vue/cli-shared-utils
      Depends on vulnerable versions of copy-webpack-plugin
      Depends on vulnerable versions of cssnano
      Depends on vulnerable versions of globby
      Depends on vulnerable versions of webpack-dev-server
    node_modules/@vue/cli-service
      @vue/cli-plugin-babel 3.4.0 - 4.5.13
        Depends on vulnerable versions of @vue/cli-service
        Depends on vulnerable versions of @vue/cli-shared-utils
      node_modules/@vue/cli-plugin-babel
        @vue/cli-plugin-router <=4.5.13
          Depends on vulnerable versions of @vue/cli-service
        node_modules/@vue/cli-plugin-router
          @vue/cli-plugin-typescript <=5.0.0-alpha.1
            Depends on vulnerable versions of @vue/cli-service
            Depends on vulnerable versions of @vue/cli-shared-utils
            Depends on vulnerable versions of fork-ts-checker-webpack-plugin
            Depends on vulnerable versions of globby
          node_modules/@vue/cli-plugin-typescript
            @vue/cli-plugin-vuex <=4.5.13
              Depends on vulnerable versions of @vue/cli-service
            node_modules/@vue/cli-plugin-vuex
```

npm audit

Security

OWASP Dependency-Track



Codebase



SBOM Tool



SBOMs
(CycloneDX format)

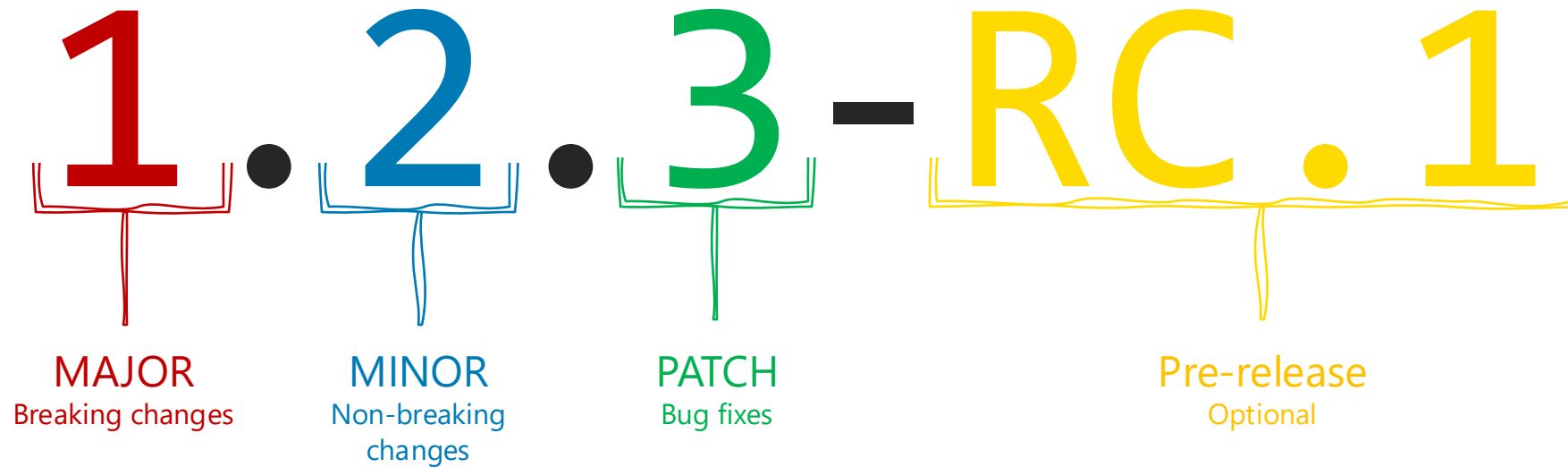


OWASP Dependency-Track

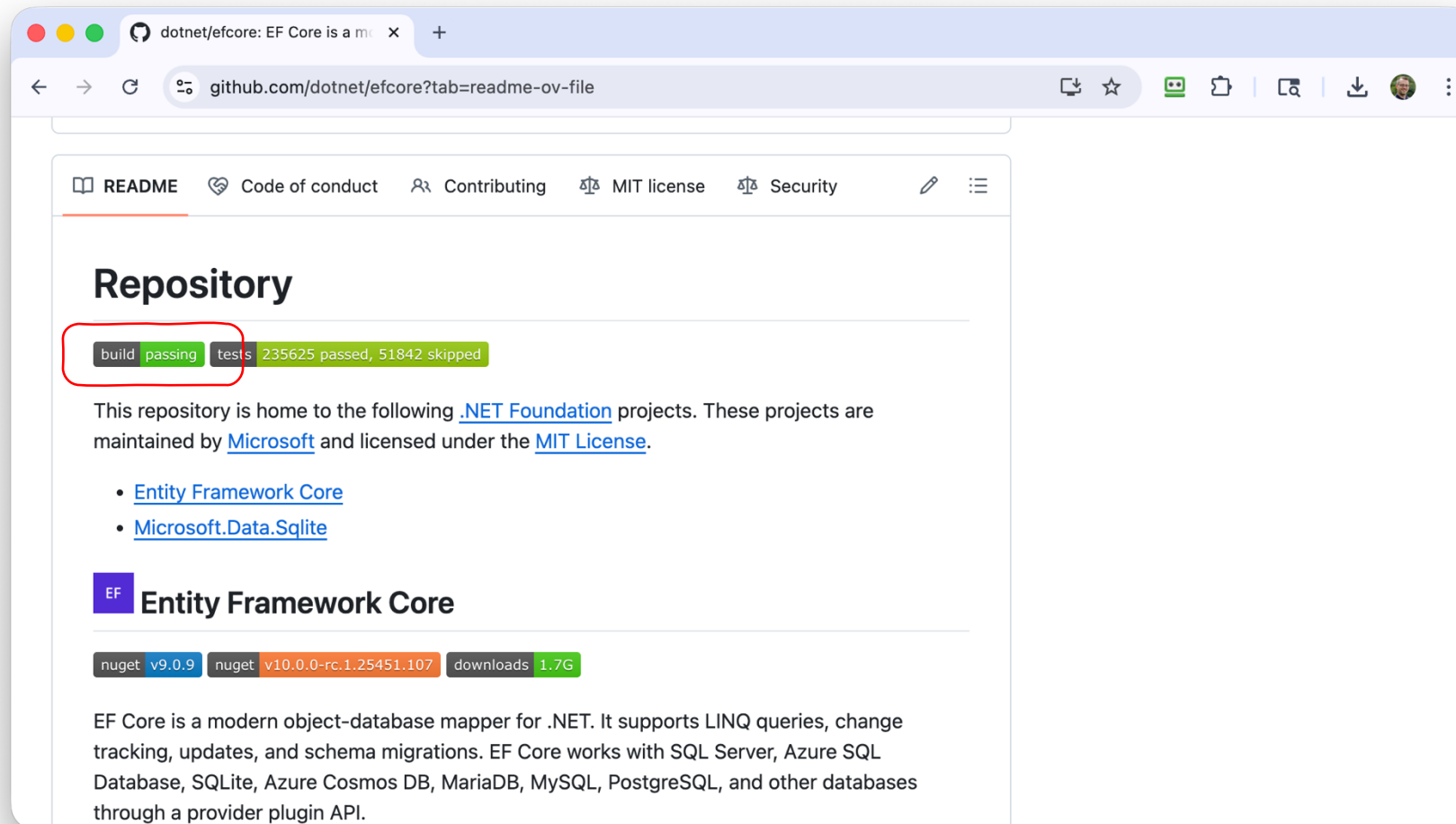


TRAILHEAD
TECHNOLOGY PARTNERS

Stability & Testing

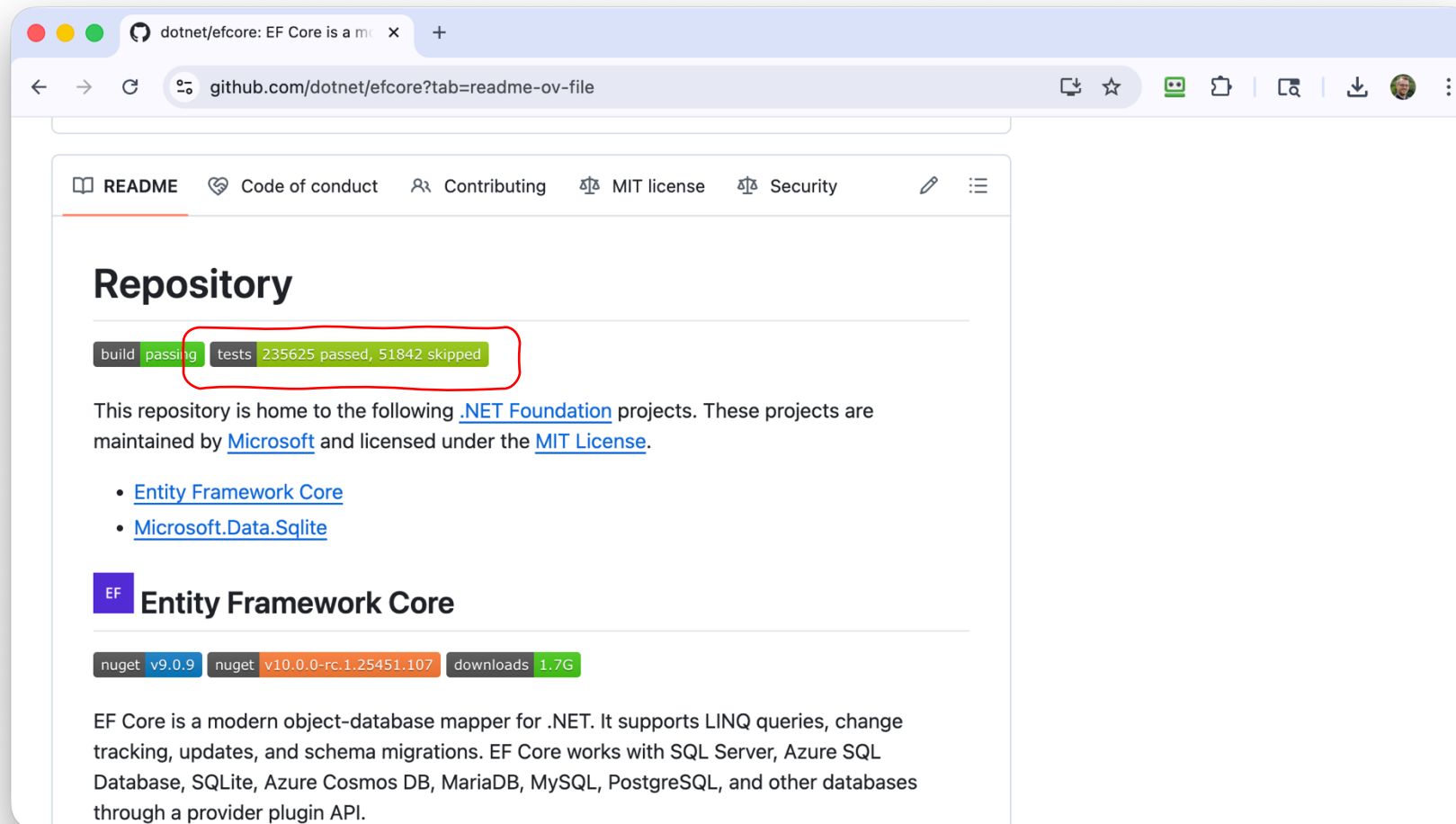


Stability & Testing



The screenshot shows the GitHub repository page for `dotnet/efcore`. The browser address bar displays `github.com/dotnet/efcore?tab=readme-ov-file`. The repository navigation bar includes links for `README`, `Code of conduct`, `Contributing`, `MIT license`, and `Security`. The `Repository` section features a status bar with a red box highlighting the `build` status as `passing` and the `tests` status as `235625 passed, 51842 skipped`. Below this, a paragraph states: "This repository is home to the following [.NET Foundation](#) projects. These projects are maintained by [Microsoft](#) and licensed under the [MIT License](#)." A bulleted list follows, containing [Entity Framework Core](#) and [Microsoft.Data.Sqlite](#). The `Entity Framework Core` section includes a status bar with `nuget v9.0.9`, `nuget v10.0.0-rc.1.25451.107`, and `downloads 1.7G`. The description text reads: "EF Core is a modern object-database mapper for .NET. It supports LINQ queries, change tracking, updates, and schema migrations. EF Core works with SQL Server, Azure SQL Database, SQLite, Azure Cosmos DB, MariaDB, MySQL, PostgreSQL, and other databases through a provider plugin API."

Stability & Testing

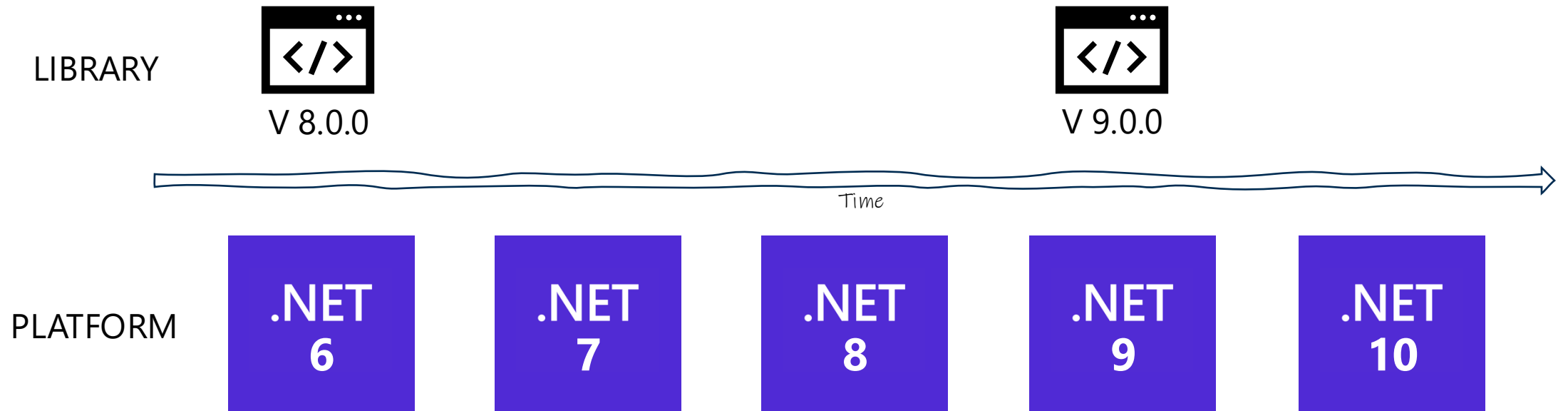


The screenshot shows the GitHub repository page for `dotnet/efcore`. The browser address bar displays `github.com/dotnet/efcore?tab=readme-ov-file`. The repository navigation bar includes links for `README`, `Code of conduct`, `Contributing`, `MIT license`, and `Security`. The `Repository` section features a status bar with a red box highlighting the build and test results: `build passing` and `tests 235625 passed, 51842 skipped`. Below this, a paragraph states: "This repository is home to the following [.NET Foundation](#) projects. These projects are maintained by [Microsoft](#) and licensed under the [MIT License](#)." A bulleted list follows, containing [Entity Framework Core](#) and [Microsoft.Data.Sqlite](#). The `Entity Framework Core` section includes a status bar with `nuget v9.0.9`, `nuget v10.0.0-rc.1.25451.107`, and `downloads 1.7G`. The description text reads: "EF Core is a modern object-database mapper for .NET. It supports LINQ queries, change tracking, updates, and schema migrations. EF Core works with SQL Server, Azure SQL Database, SQLite, Azure Cosmos DB, MariaDB, MySQL, PostgreSQL, and other databases through a provider plugin API."

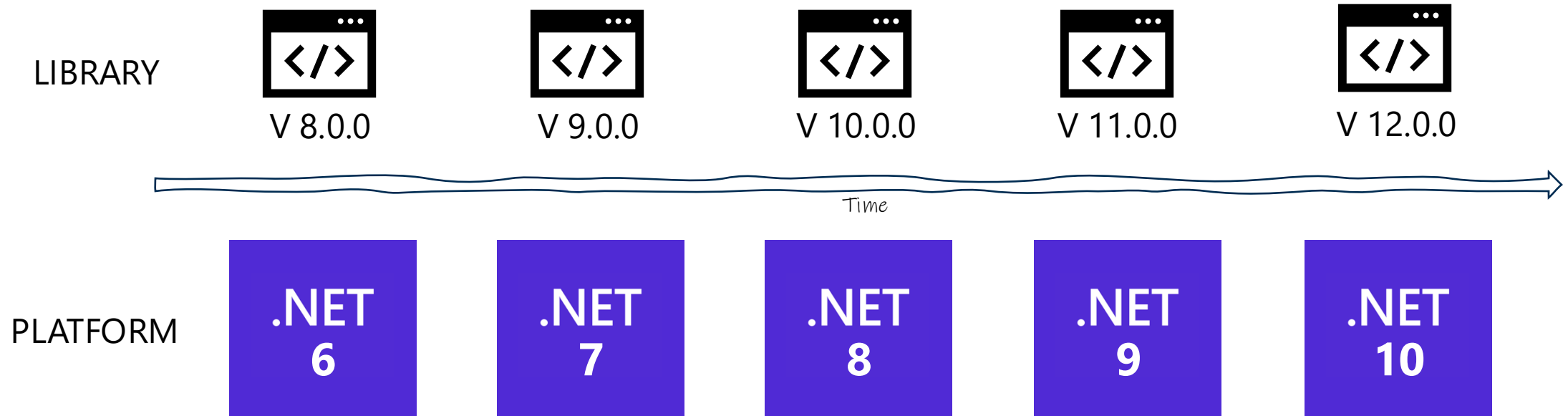
Fit for Purpose

Does it play nice with our stack?

Platform Alignment



Platform Alignment



Platform Alignment

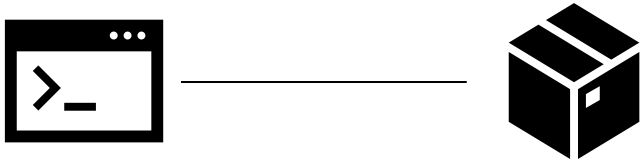
Third-Party Library

.NET Standard

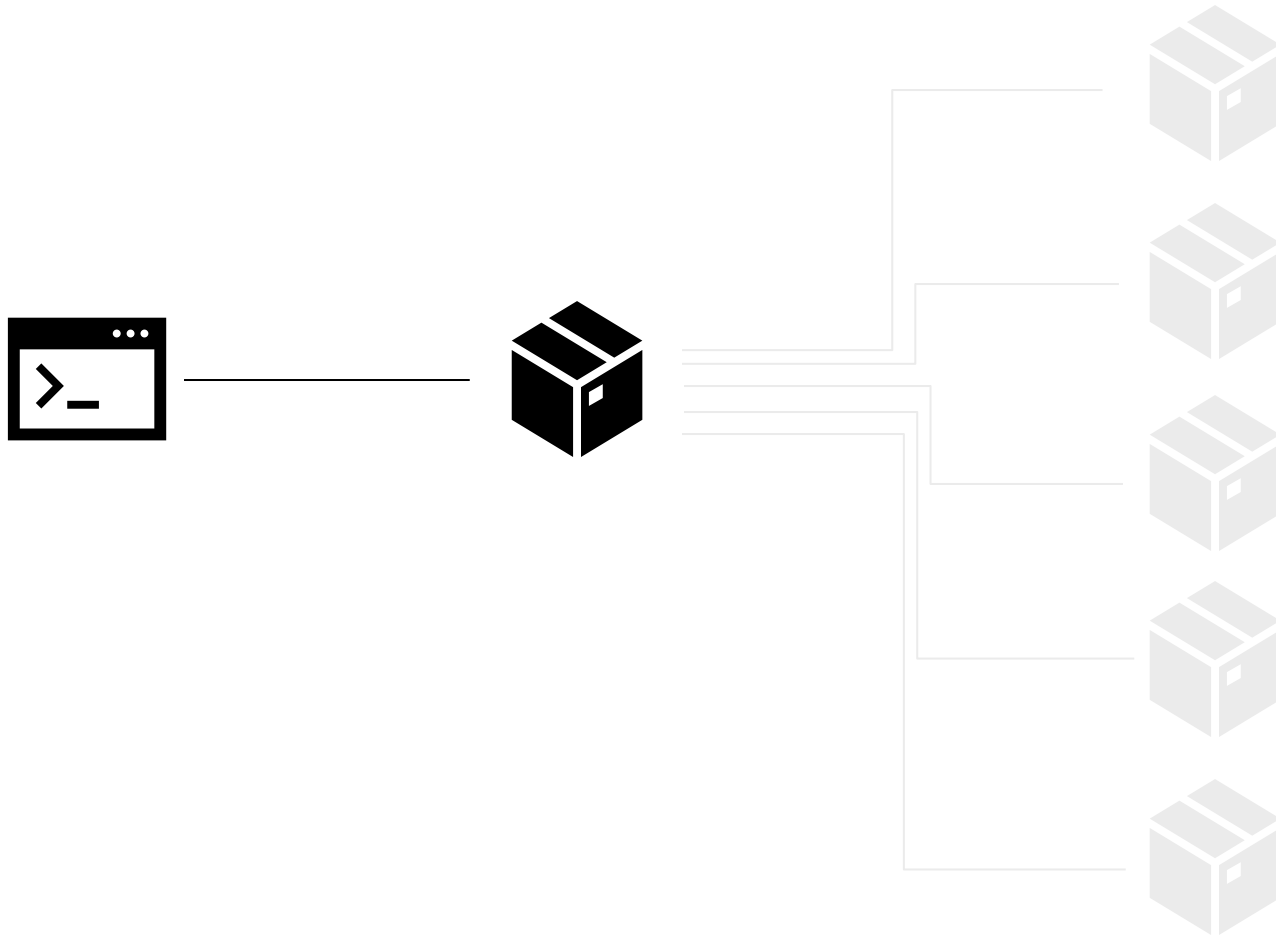
.NET
Framework

.NET

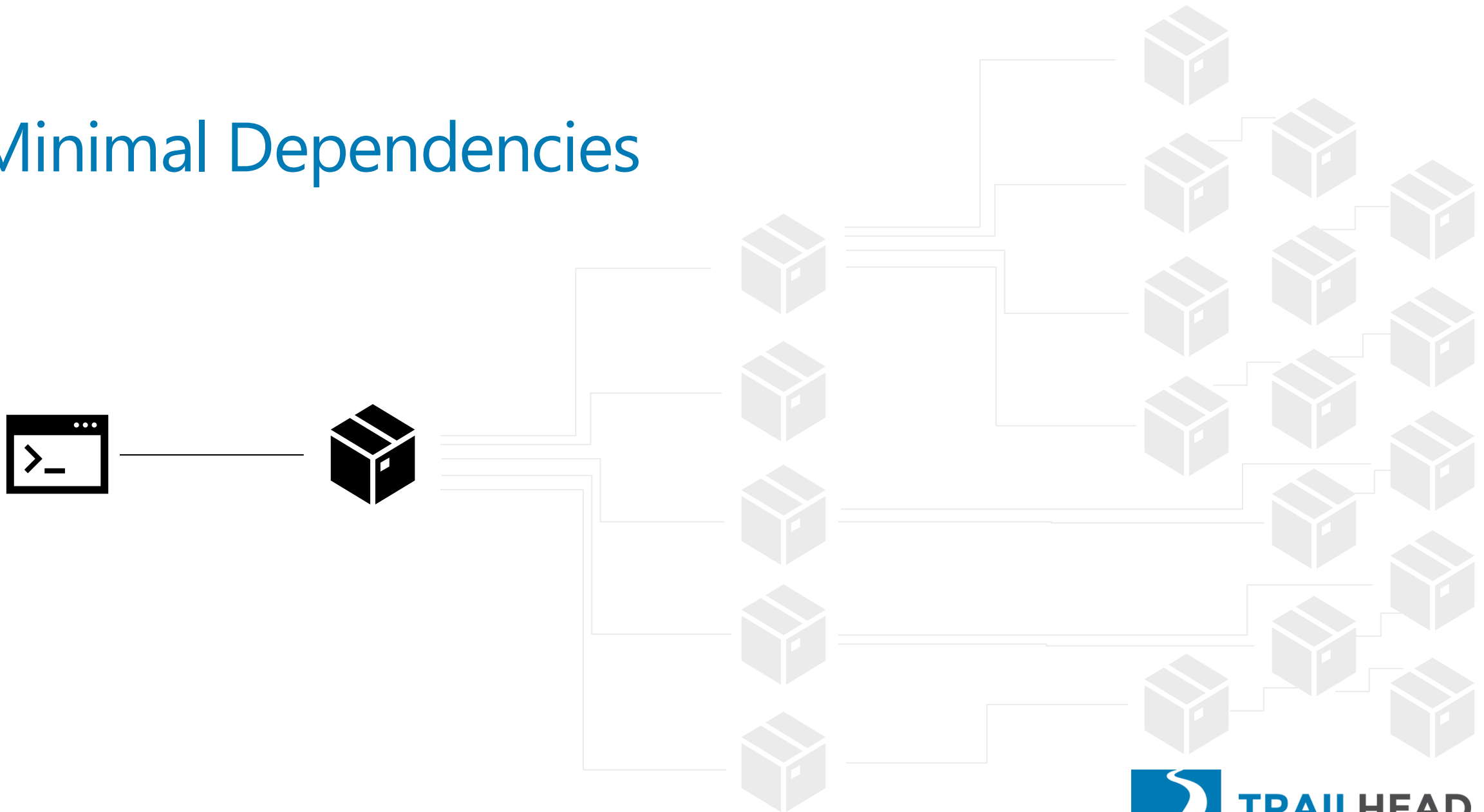
Minimal Dependencies



Minimal Dependencies



Minimal Dependencies



Right-Sized for the Job



Does **One Job**,
And Does It **Well**

Right-Sized for the Job



Feature Set Is
Not **Overkill**

Right-Sized for the Job

Features	Necessary	Cohesive	Not Avail Elsewhere
Library Feature 1	✓	✓	✓
Library Feature 2	✓	✓	✓
Library Feature 3	✓	✓	✓
Library Feature 4	✓	✓	✓
Library Feature 5	✓	✓	✓
Library Feature 6	✓	✓	✓
Library Feature 7	✓	✓	✓

Right-Sized for the Job

Features	Necessary	Cohesive	Not Avail Elsewhere
Library Feature 1	✗	✓	✓
Library Feature 2	✓	✓	✓
Library Feature 3	✗	✗	✗
Library Feature 4	✓	✓	✓
Library Feature 5	✓	✓	✓
Library Feature 6	✓	✓	✗
Library Feature 7	✓	✗	✗

Right-Sized for the Job

Features	Necessary	Cohesive	Not Avail Elsewhere
Library Feature 1	✓	✗	✓
Library Feature 2	✓	✗	✓
Library Feature 3	✗	✗	✗
Library Feature 4	✗	✗	✗
Library Feature 5	✗	✓	✓
Library Feature 6	✗	✗	✗
Library Feature 7	✗	✗	✗

Going Deeper

The Rules You *Might Not* Be Thinking About



Mattias Karlsson (he/him)

@devlead

Follow



One way to avoid vendor lock-in is to embrace all that the vendor has to offer, but ensure things are small, isolated, uncomplicated, and replaceable.

7:41 AM · Jul 21, 2022



TRAILHEAD
TECHNOLOGY PARTNERS



Mattias Karlsson (he/him)

@devlead

Follow



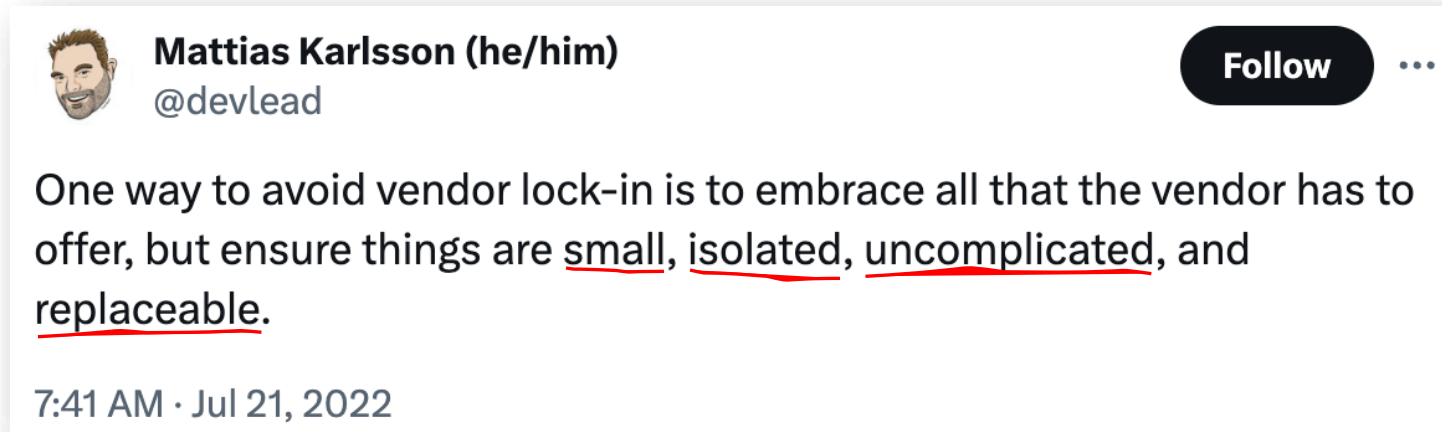
One way to avoid vendor lock-in is to embrace all that the vendor has to offer, but ensure things are small, isolated, uncomplicated, and replaceable.

7:41 AM · Jul 21, 2022



TRAILHEAD
TECHNOLOGY PARTNERS

1. Small
2. Isolated



3. Uncomplicated
4. Replaceable

1. Small
2. Isolated



Mattias Karlsson (he/him)

@devlead

Follow



These rules don't *just* avoid vendor lock-in...

One way to avoid vendor lock-in is to ensure that your vendor has to offer, but ensure things are small, isolated, uncomplicated, and replaceable.

7:41 AM · Jul 21, 2022

3. Uncomplicated
4. Replaceable



TRAILHEAD
TECHNOLOGY PARTNERS

1. Small
2. Isolated



Mattias Karlsson (he/him)

@devlead

Follow



**they ensure great 3rd-party
libraries *in general*!**

7:41 AM · Jul 21, 2022

3. Uncomplicated
4. Replaceable



TRAILHEAD
TECHNOLOGY PARTNERS

So Let's Talk About These...

1. Small



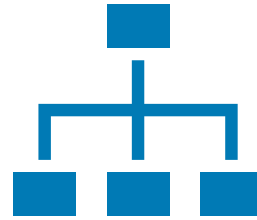
What “Small” Means

It does **one thing** well without
unnecessary bloat

Why Small Is Better



Less to learn



Fewer dependencies



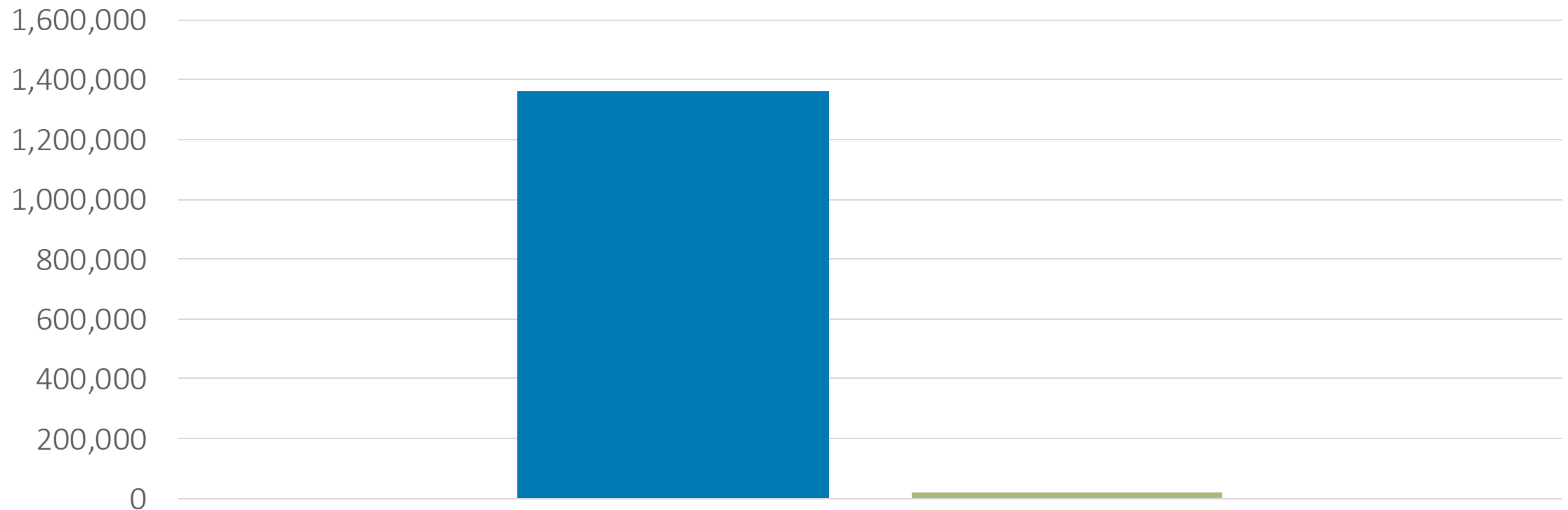
Easier upkeep

Case Study

Entity Framework



Dapper



Lines Of Code (less is better)

■ EF Core ■ Dapper

Case Study

Entity Framework



Dapper



Case Study

Entity Framework



Dapper

Feature	EF Core	Dapper
Object Mapping	✓	✓
Query Generation	✓	✗
Change Tracking	✓	✗
Database Migrations	✓	✗
Lazy Loading	✓	✗
Transaction Management	✓	✗
Concurrency Control	✓	✗
Caching	✓	✗

2. Isolated



What “Isolated” Means

It has **minimal external dependencies** and doesn't **tightly couple** you to its ecosystem

Why Isolated Is Better



Lower Risk of
Dependency Hell



Lower Attack Surface



Predictable Behavior

Case Study

Entity Framework



Dapper

```
var myOrders = dbContext.Orders.ToList(); // A: IEnumerable
```

```
var myOrders = dbContext.Orders; // B: IQueryable
```

```
// Thousands of lines of code...
```

```
// What will happen here?
```

```
var order = myOrders
```

```
    .FirstOrDefault(o => o.OrderID == 170);
```



TRAILHEAD
TECHNOLOGY PARTNERS

Case Study

Entity Framework



Dapper

```
var myOrders = dbContext.Orders.ToList(); // A: IEnumerable
```

```
var myOrders = dbContext.Orders; // B: IQueryable
```

```
// Thousands of lines of code...
```

```
// What will happen here?
```

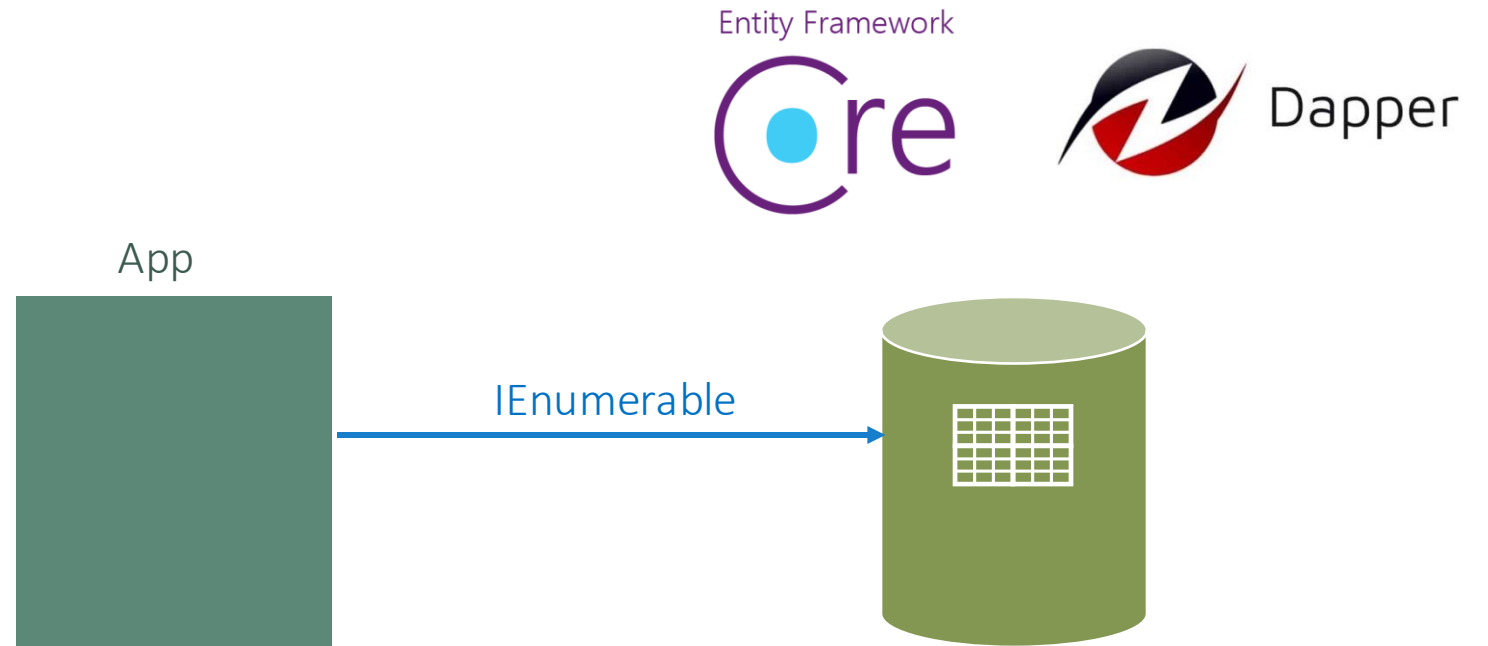
```
var order = myOrders  
    .FirstOrDefault(o => o.OrderID == 170);
```



TRAILHEAD
TECHNOLOGY PARTNERS

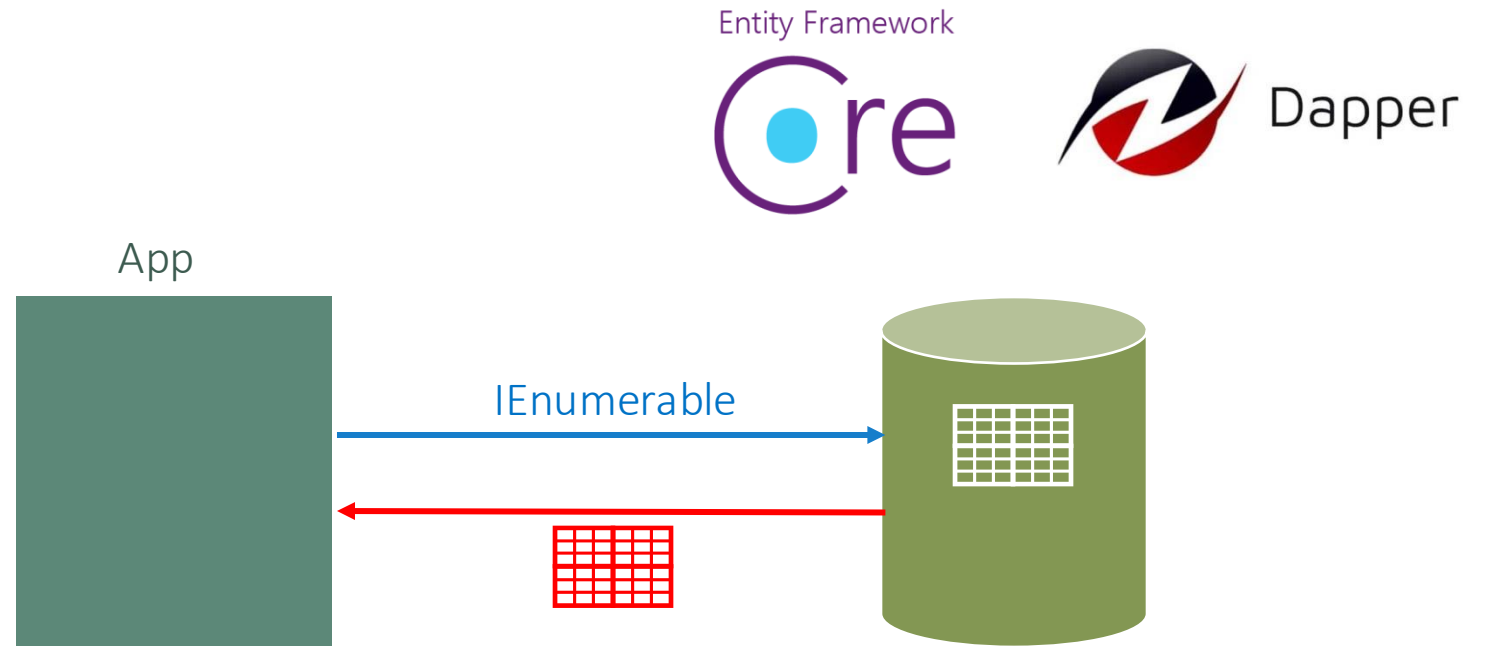
Case Study

IEnumerable



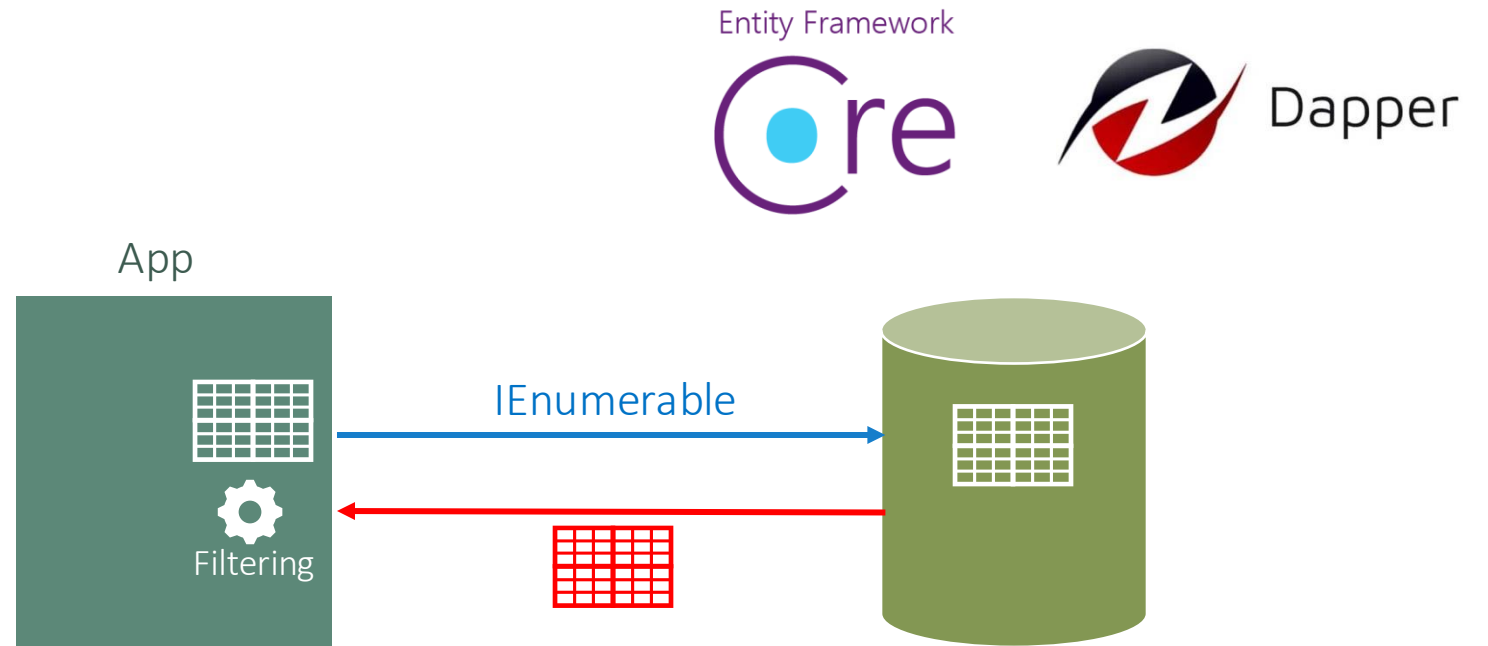
Case Study

IEnumerable



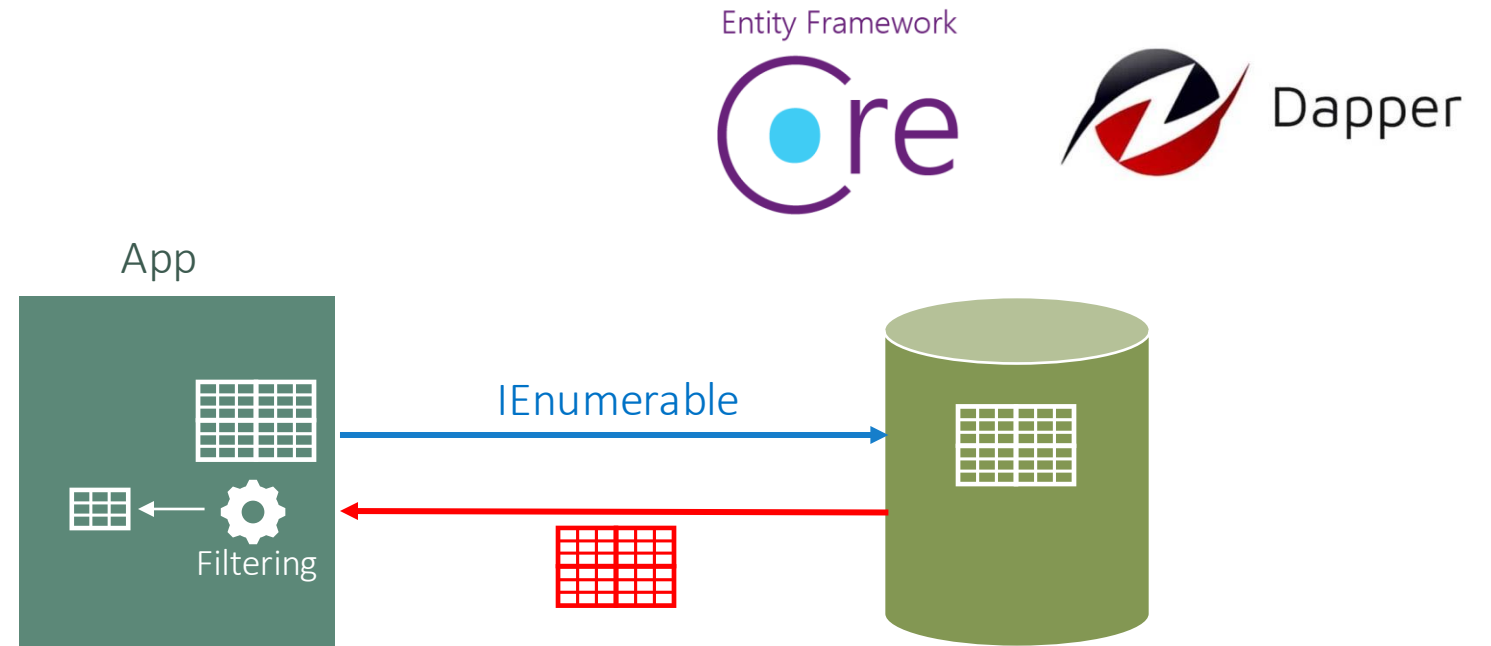
Case Study

IEnumerable



Case Study

IEnumerable



Case Study

Entity Framework



Dapper

```
var myOrders = dbContext.Orders.ToList(); // A: IEnumerable
```

```
var myOrders = dbContext.Orders; // B: IQueryable
```

```
// Thousands of lines of code...
```

```
// What will happen here?
```

```
var order = myOrders  
    .FirstOrDefault(o => o.OrderID == 170);
```



TRAILHEAD
TECHNOLOGY PARTNERS

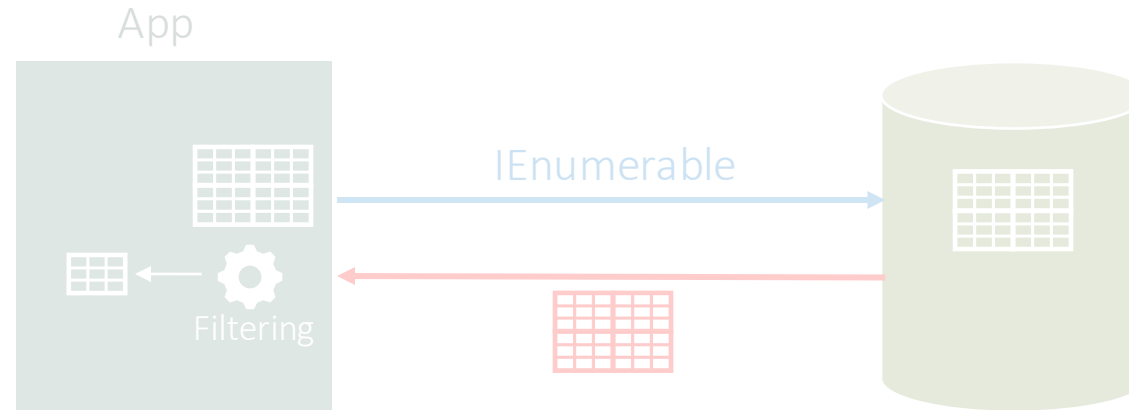
Case Study

Entity Framework

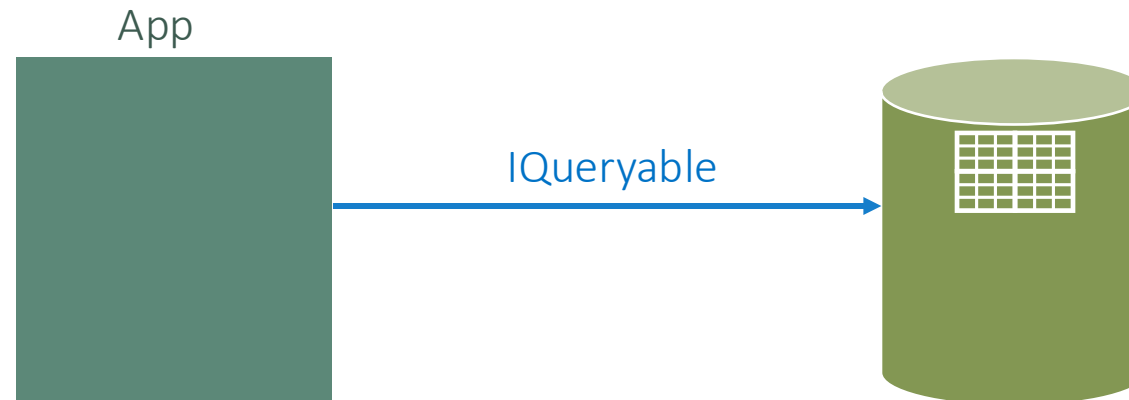


Dapper

IEnumerable



IQueryable



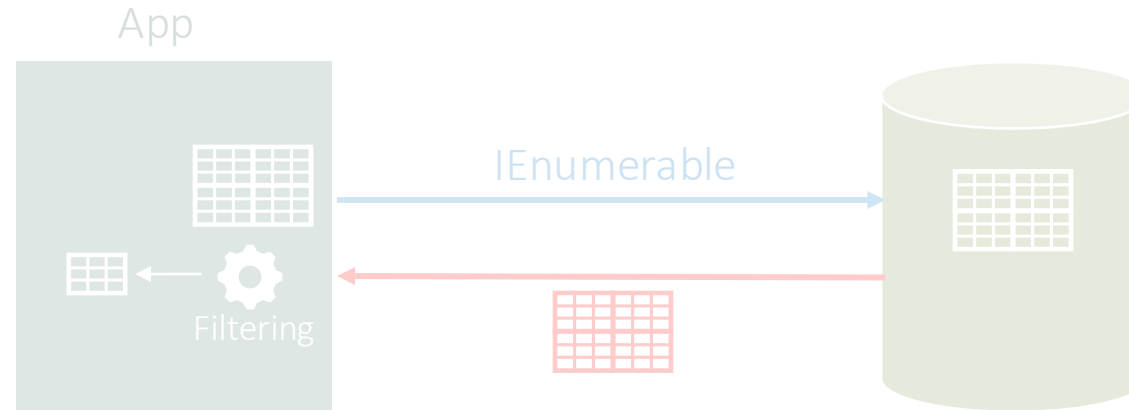
Case Study

Entity Framework

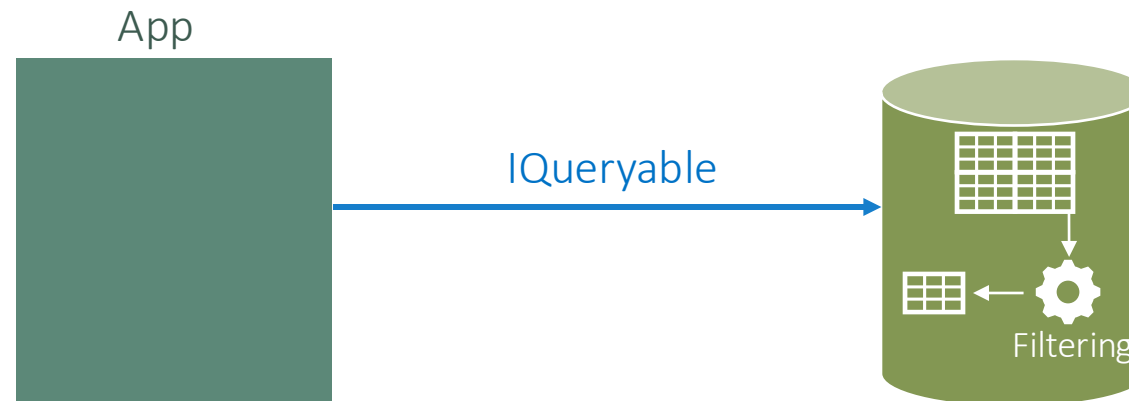


Dapper

IEnumerable



IQueryable



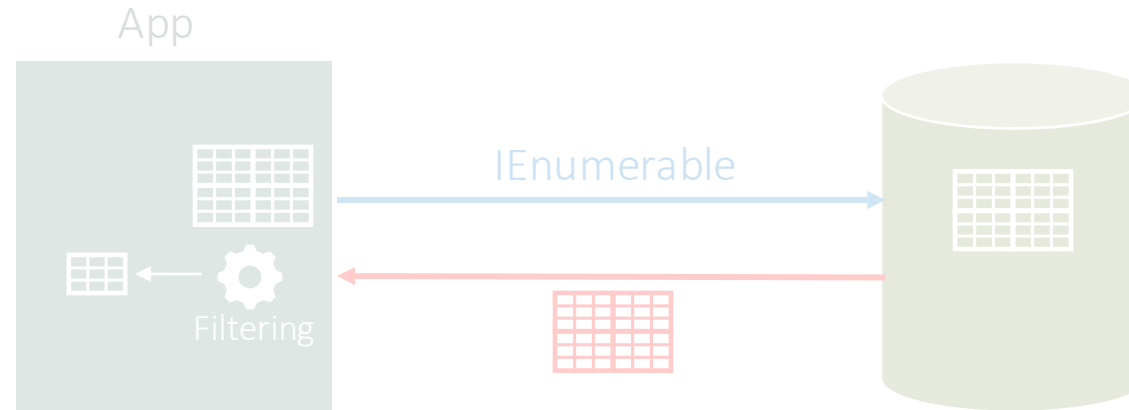
Case Study

Entity Framework

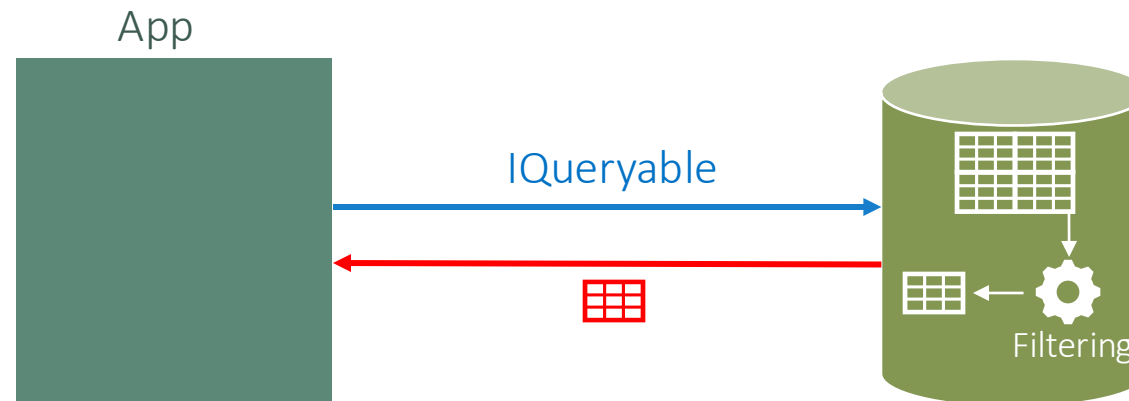


Dapper

IEnumerable



IQueryable



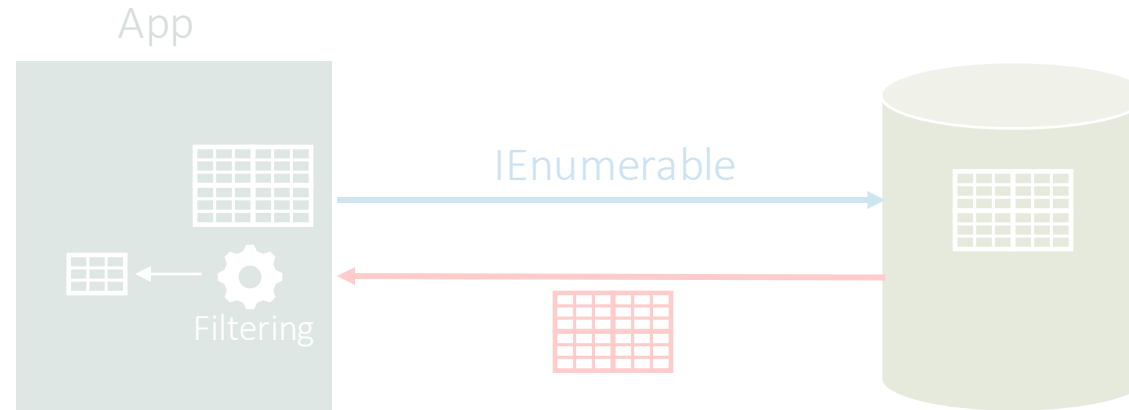
Case Study

Entity Framework

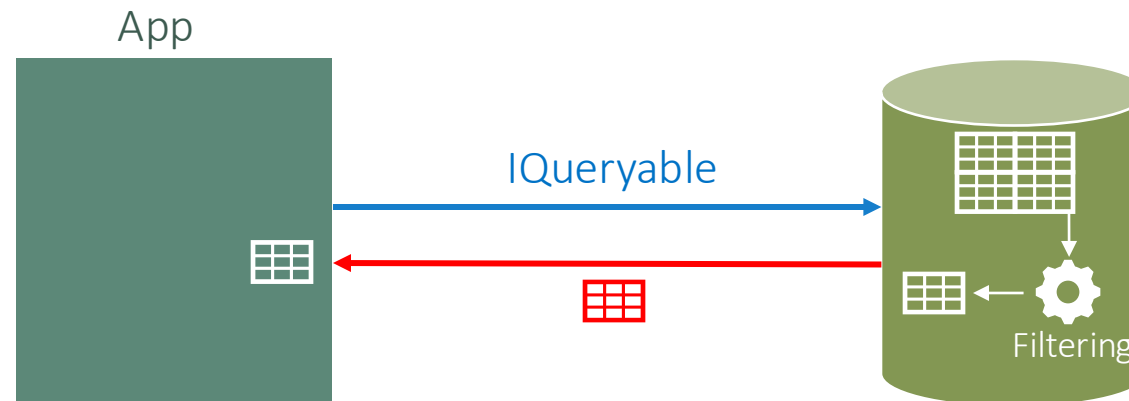


Dapper

IEnumerable



IQueryable



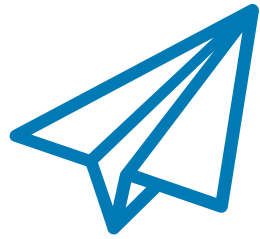
3. Uncomplicated



What “Uncomplicated” Means

It has a **clear, intuitive API**
with good **documentation**
and **predictable** behavior

Uncomplicated



Less Drag



Shallow Curve



Faster Delivery

Case Study

Entity Framework



 Docs are excellent!

 Composable queries

 Powerful when well understood



Case Study

Entity Framework



 N+1 Queries

 Forgetting to
Configure
Concurrency Handling

 Data Loss Due to
Accidental Database
Migration

 Misunderstanding
Mapping Conventions

 Beware of
Lazy Loading

 Poor
Performing
Generated
SQL

 Forgetting to Disable
Change Tracking

 Forgetting to
Use Async
Method

 Confusing
IEnumerable
and
IQueryable

 Forgetting to
Dispose of
DbContext

Dapper Case Study



```
// One query, exactly once (no deferred execution, no N+1)
const string sql = @"
    SELECT o.Id, o.Date, i.Id AS ItemId, i.Sku, i.Qty
    FROM Orders o
    JOIN OrderItems i ON i.OrderId = o.Id
    WHERE o.CustomerId = @customerId
    ORDER BY o.Date DESC";

var rows = connection.Query(sql, new { customerId });
```



Dapper Case Study



```
// Predictable updates (no change tracking)
const string sql = @"
    SELECT o.Id, o.Date, i.Id AS ItemId, i.Sku, i.Qty
    FROM Orders o
    JOIN OrderItems i ON i.OrderId = o.Id
    WHERE o.CustomerId = @customerId
    ORDER BY o.Date DESC";

var rows = connection.Query(sql, new { customerId });
```



Dapper Case Study



```
// Deterministic pagination & parameters (safe, explicit)
```

```
const string sql = @"
```

```
    SELECT Id, Name
```

```
    FROM Products
```

```
    WHERE Category = @Category
```

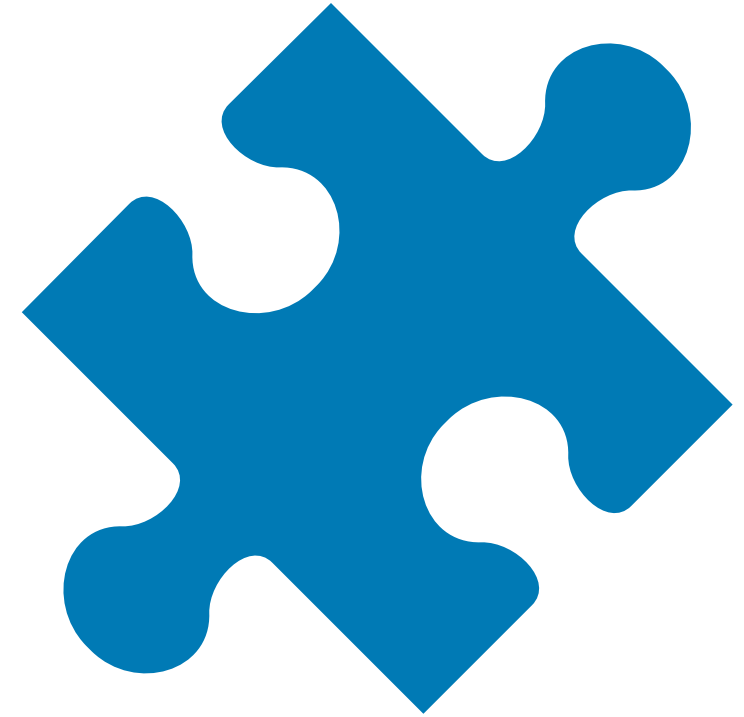
```
    ORDER BY Name
```

```
    OFFSET @Skip ROWS FETCH NEXT @Take ROWS ONLY";
```

```
var products = connection.Query<Product>(sql, new { Category = "Books", Skip = 20,  
Take = 10 });
```



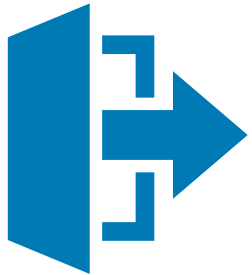
4. Replaceable



What “Replaceable” Means

It can be **swapped out** with another library **without changes leaking** into a lot of your codebase

Replaceable



Exit Ready

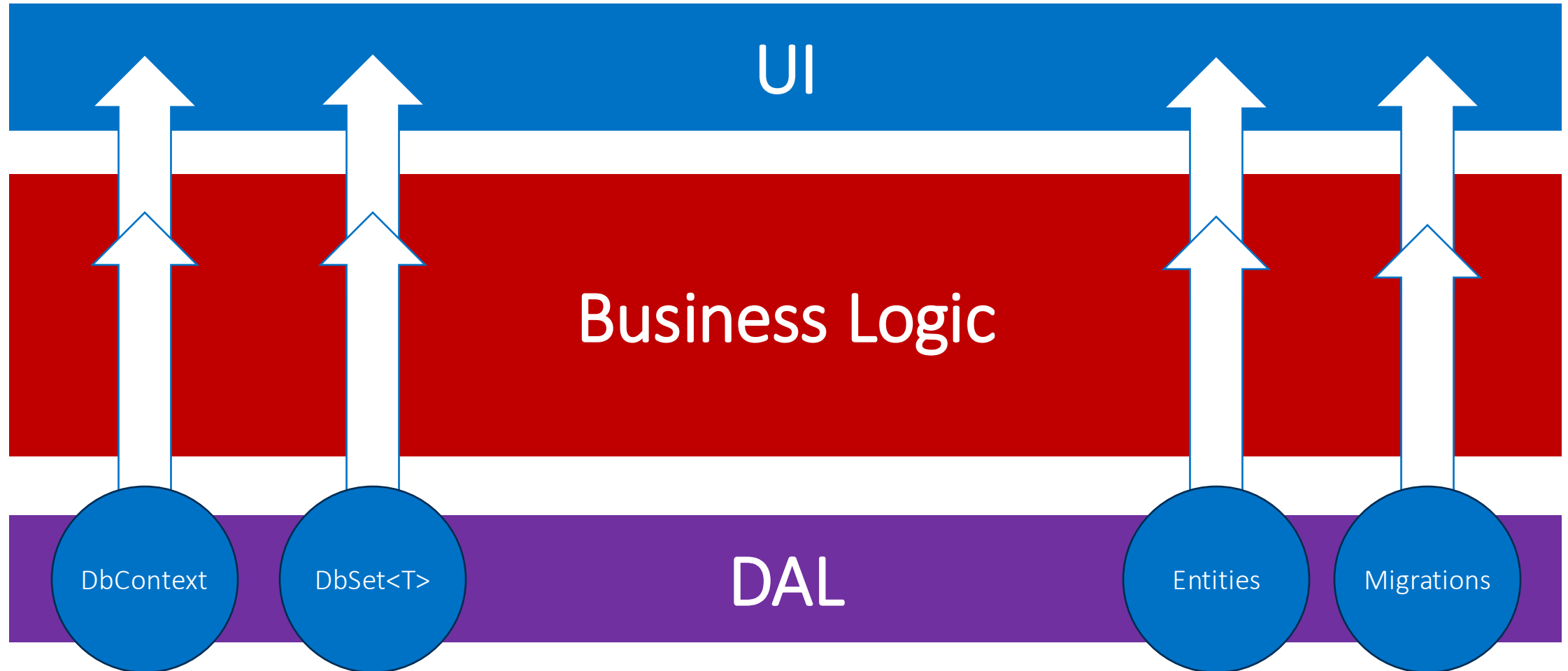


Abstracted



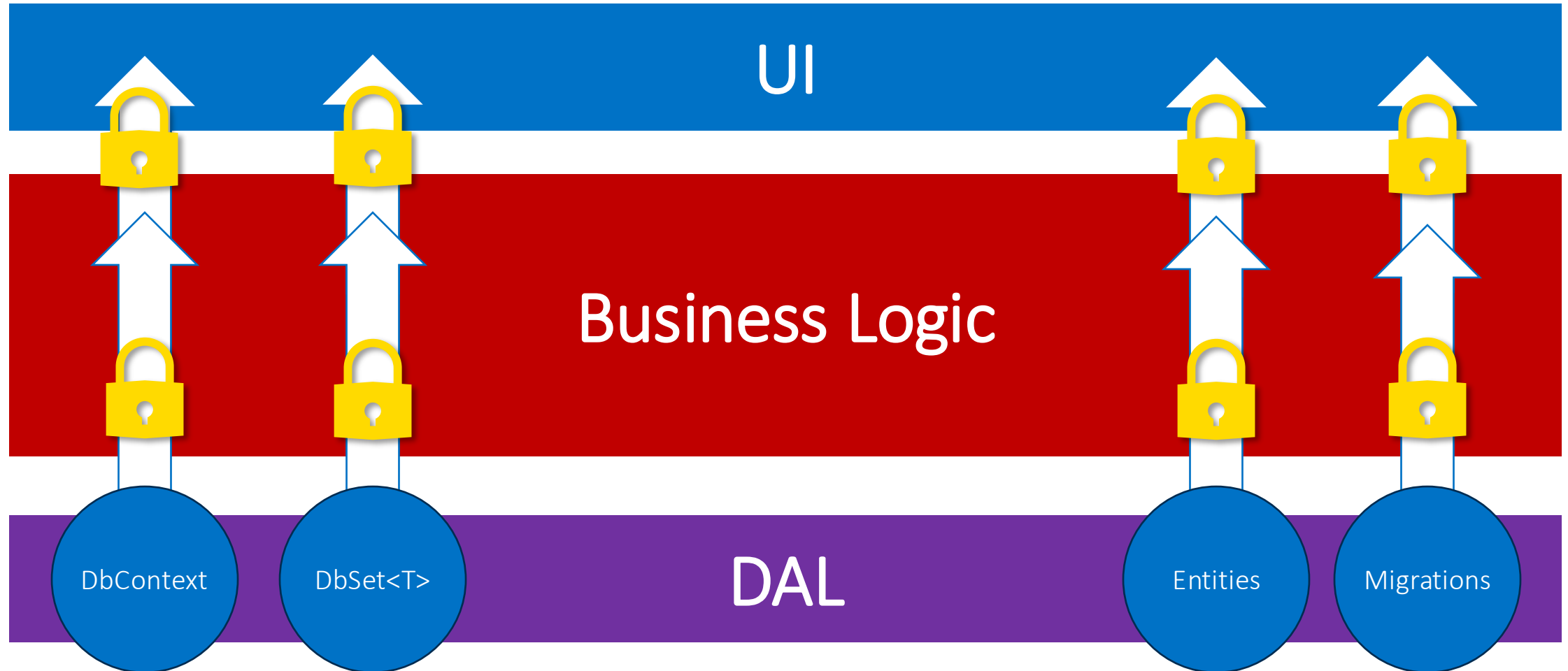
In Control

Entity Framework Case Study

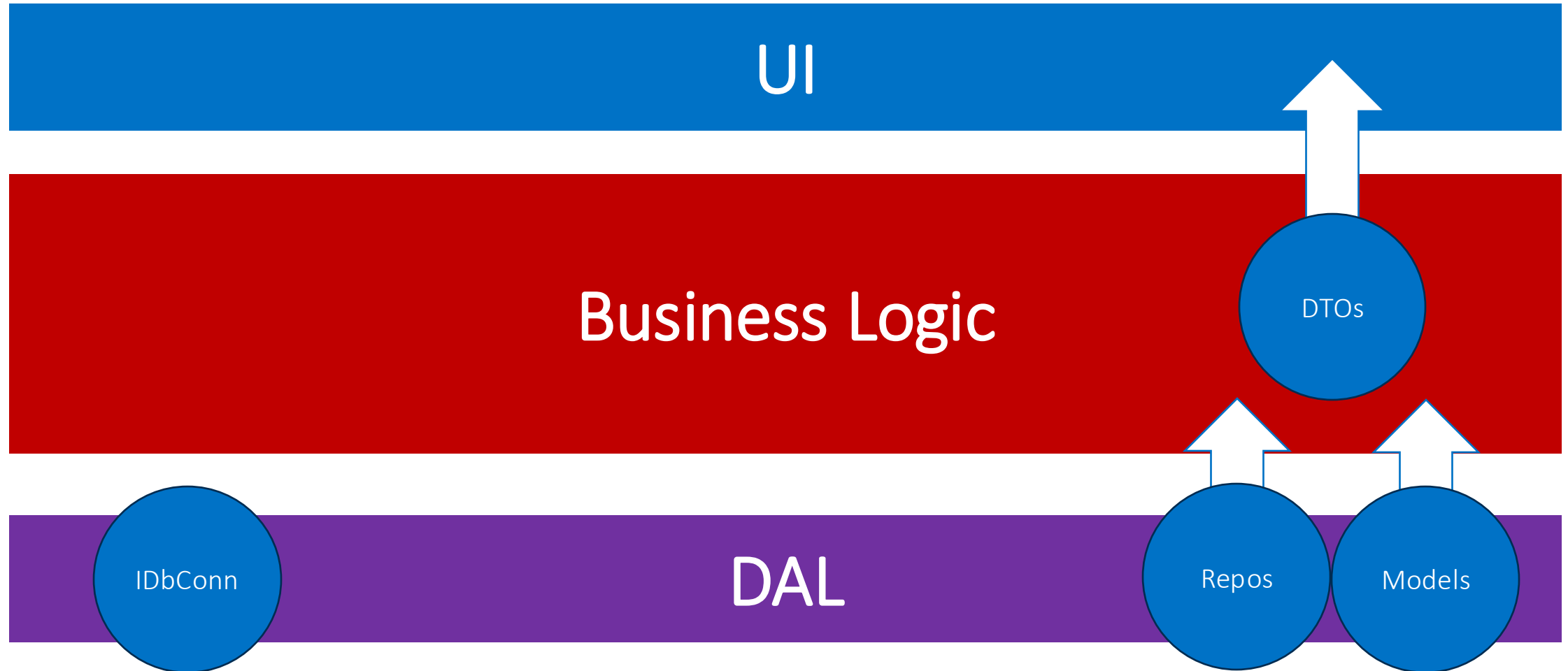


Entity Framework Case Study

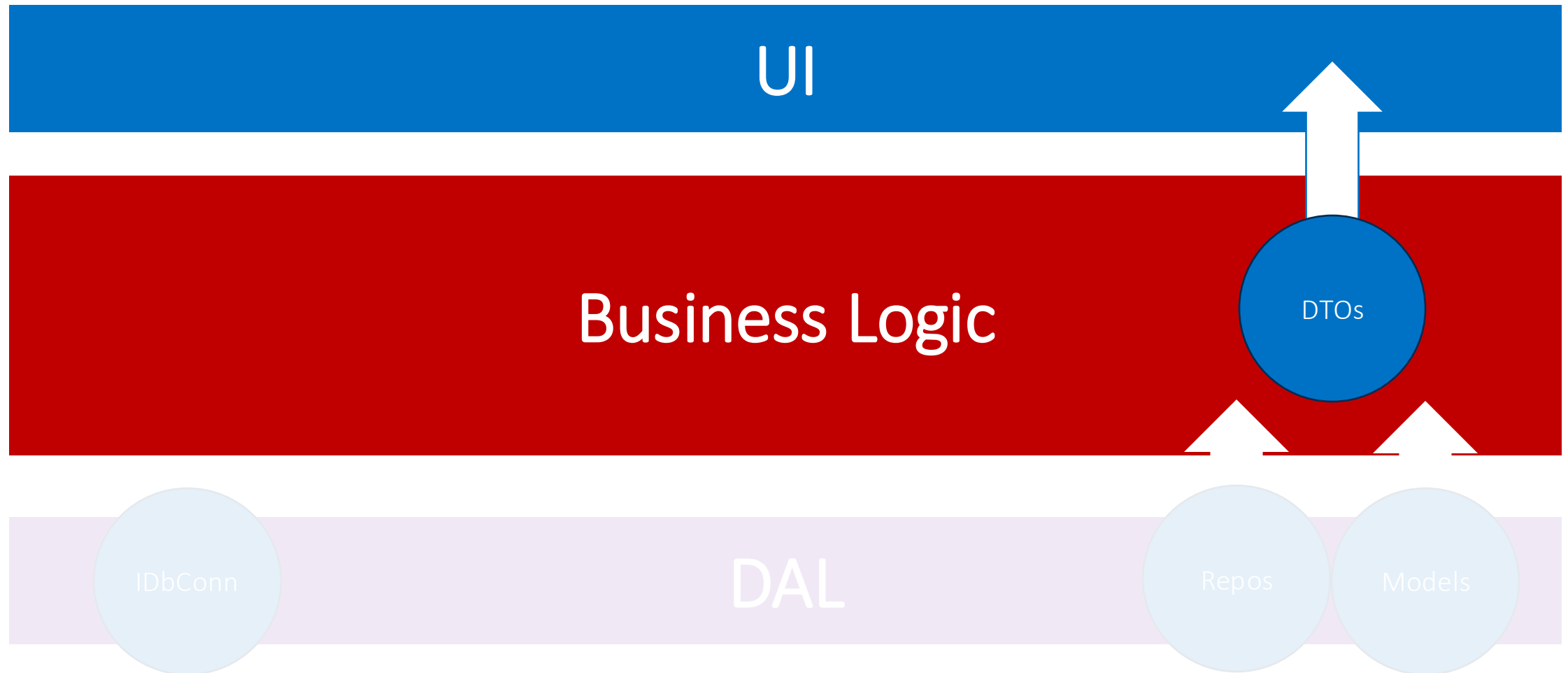
Entity Framework



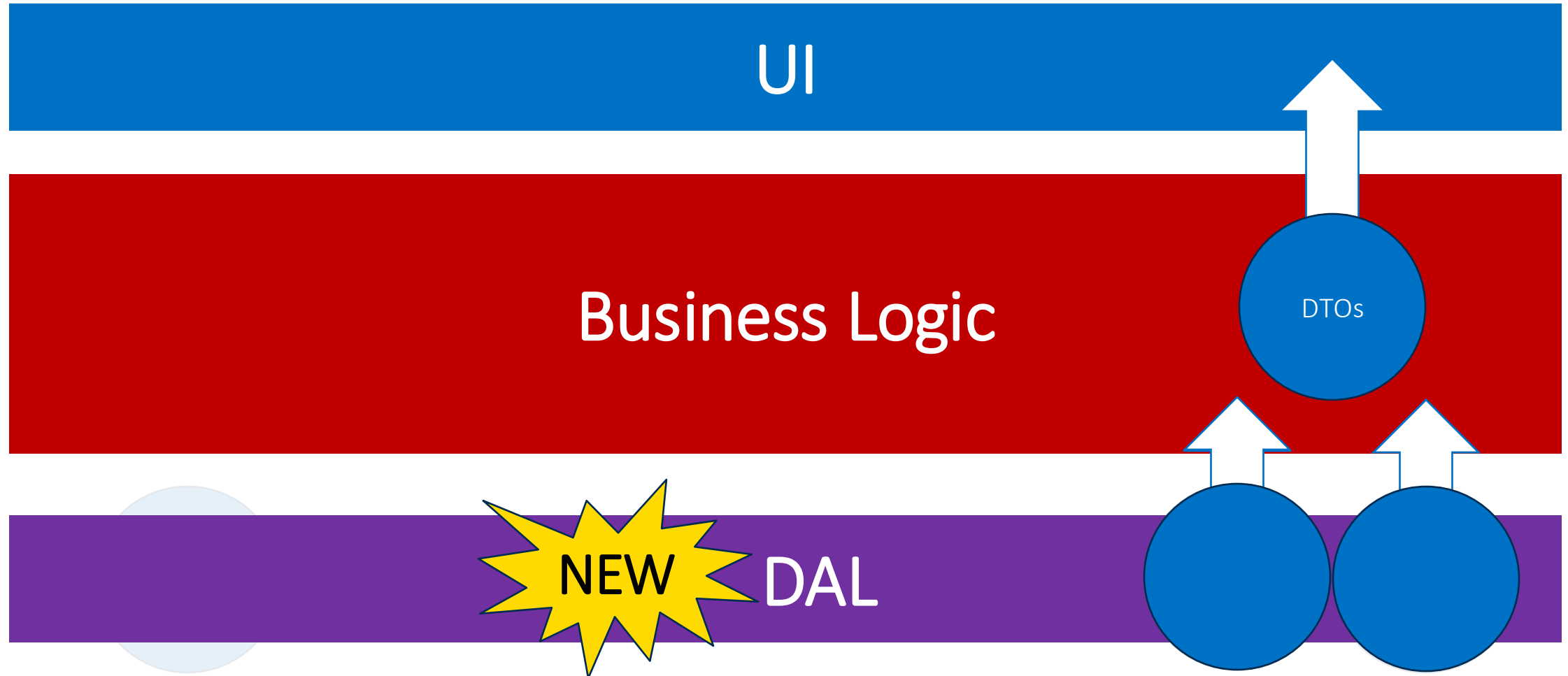
Dapper Case Study



Dapper Case Study



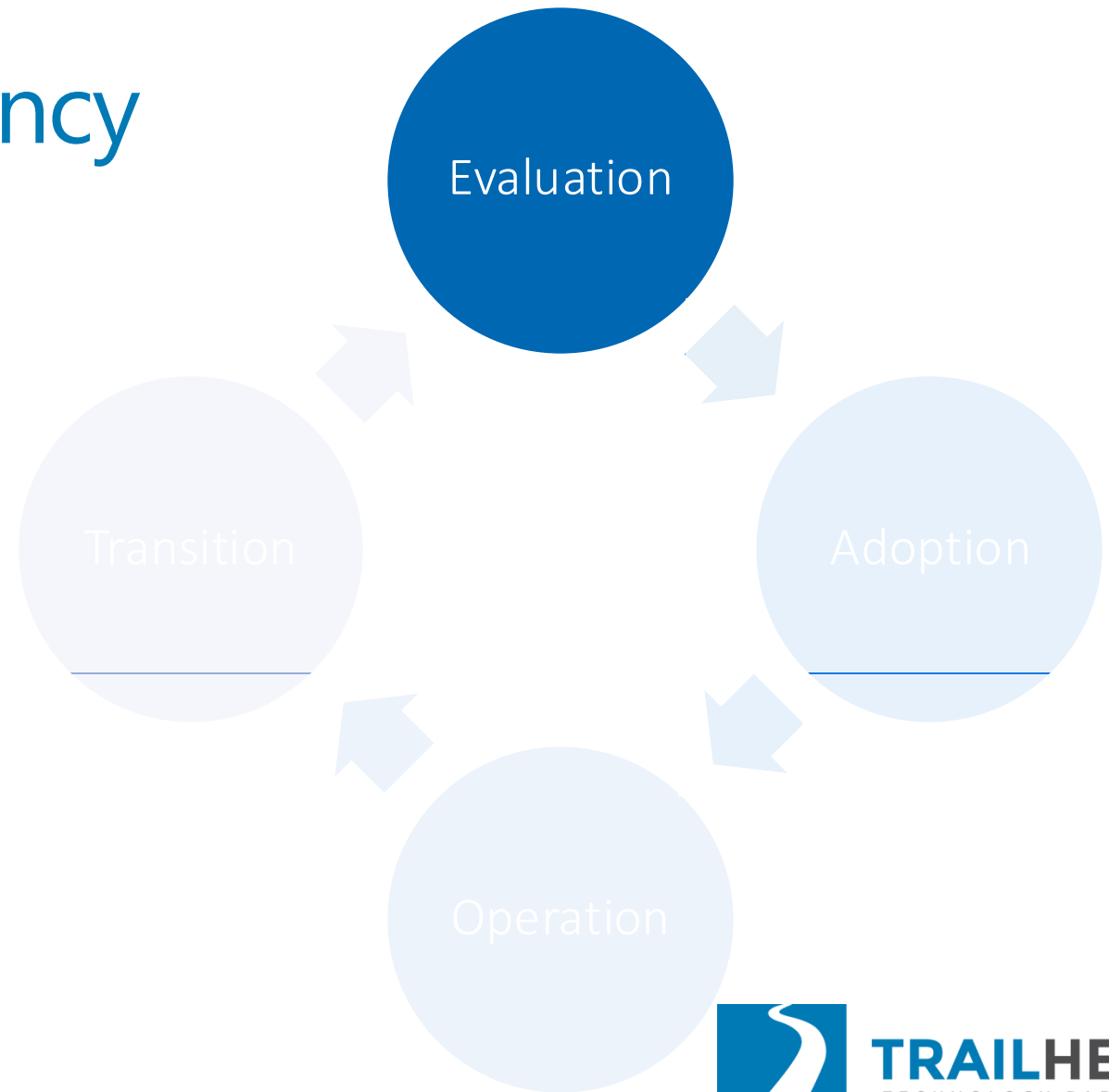
Dapper Case Study



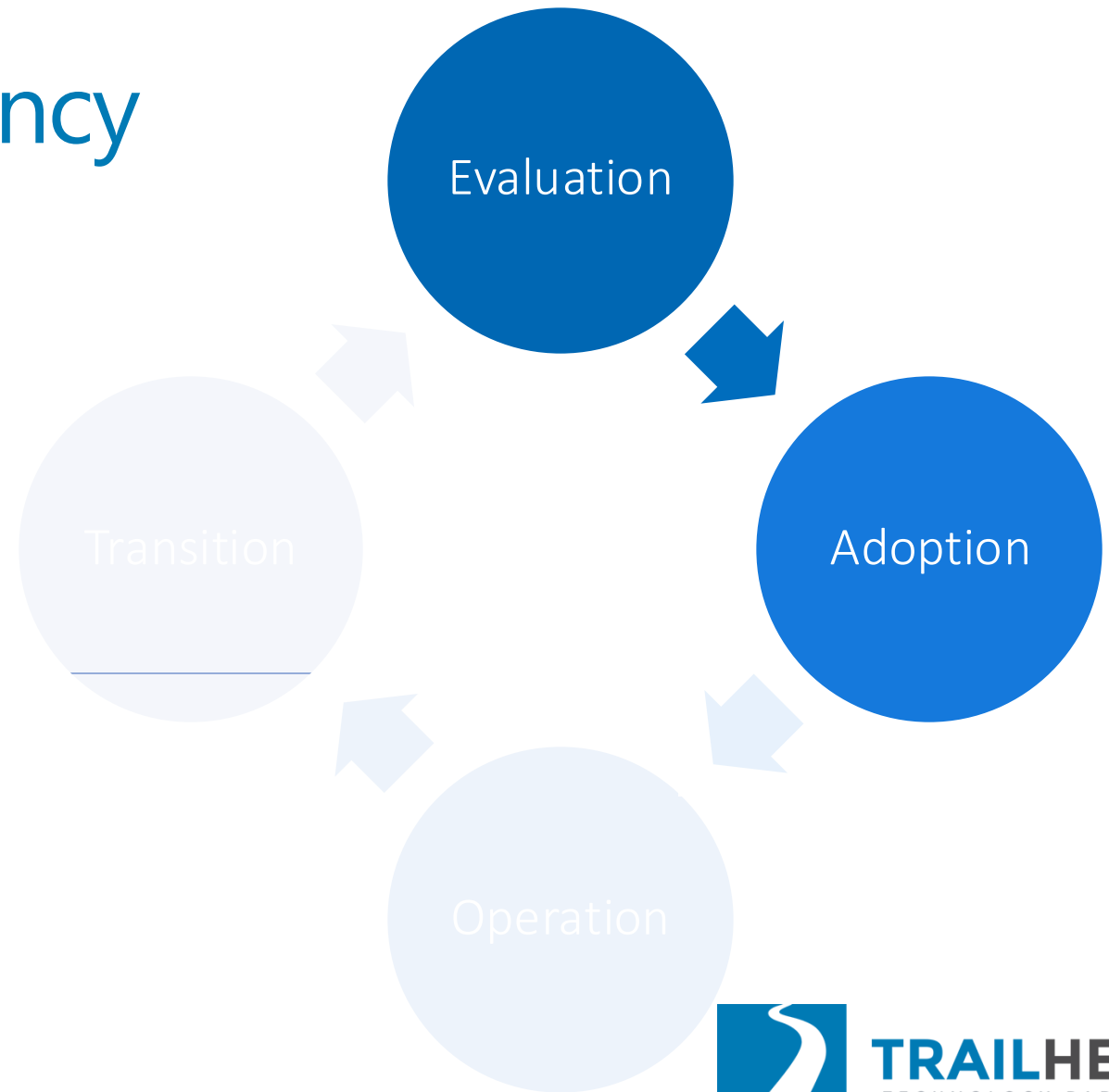
Lifecycle of a Dependency

A Checklist for Each Phase

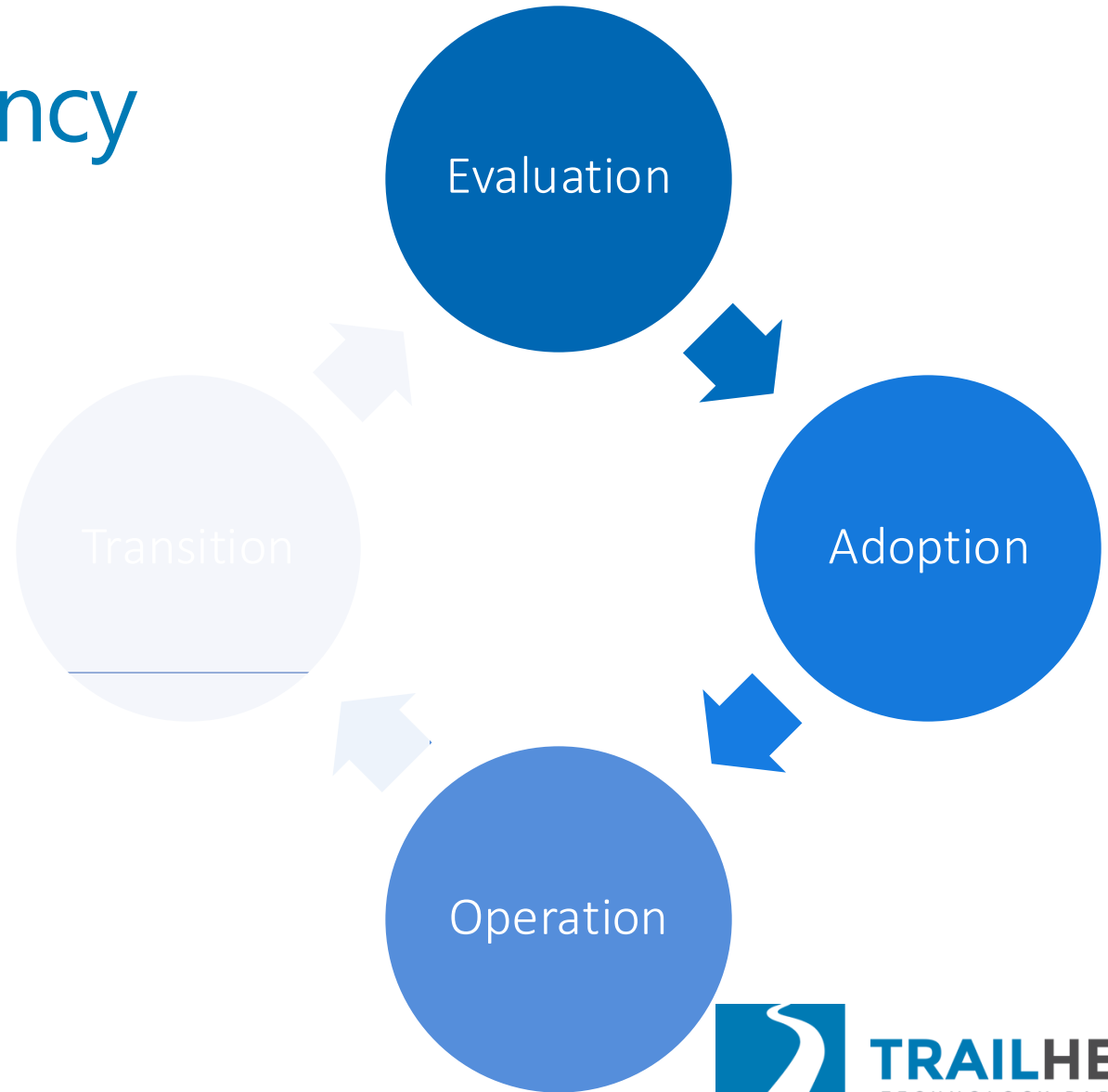
Lifecycle of a Dependency



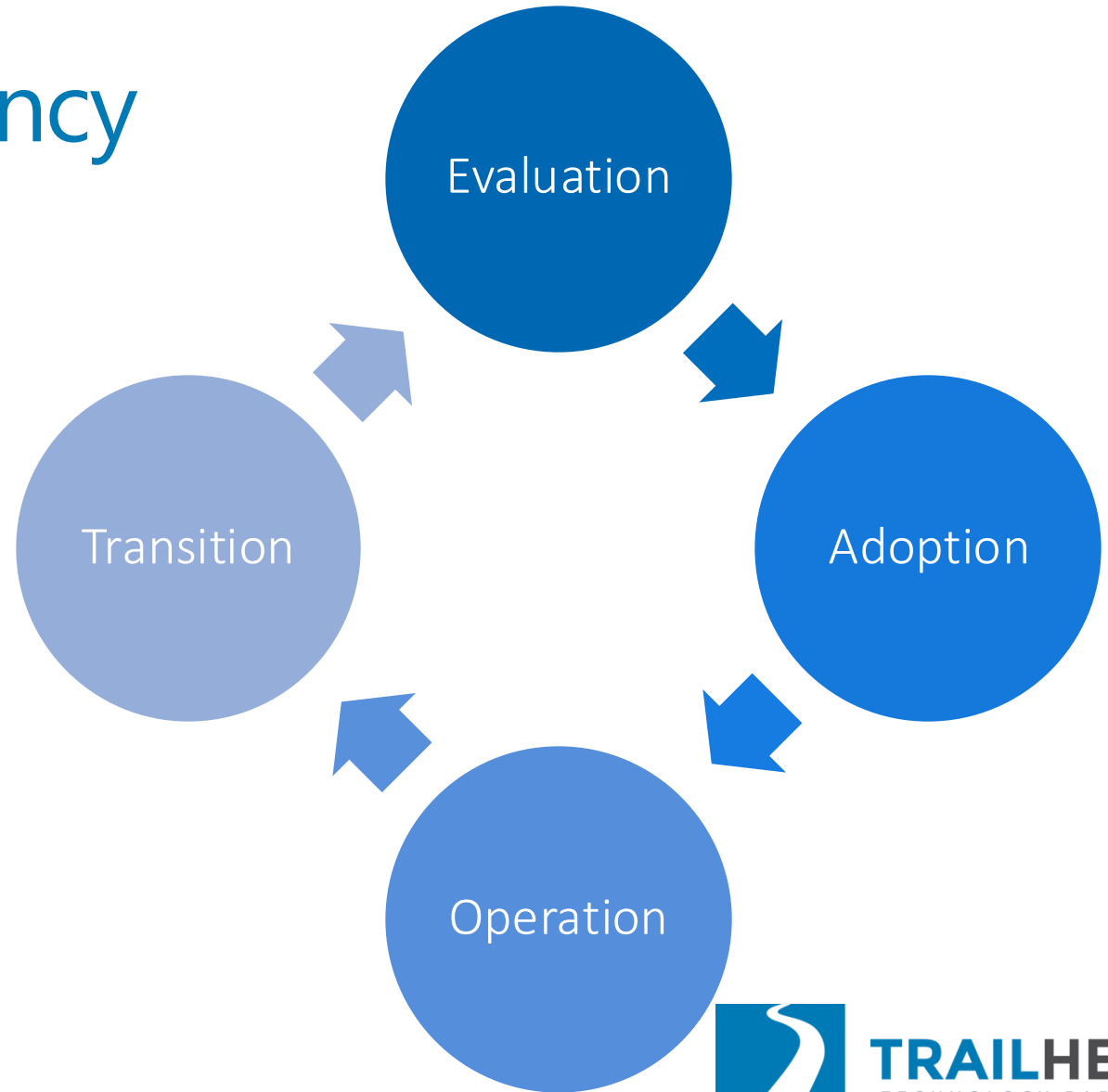
Lifecycle of a Dependency



Lifecycle of a Dependency



Lifecycle of a Dependency



How do I best **manage**
third-party libraries in
each **stage**?

Evaluation Stage Checklist

- ✓ Check project activity & health
- ✓ Review community & ecosystem
- ✓ Verify license for fit
- ✓ Run security scans of dependencies
- ✓ Consider the 4 deeper rules: **small, isolated, uncomplicated, replaceable**

Adoption Stage Checklist

- ✓ Document choices in an ADR (Architecture Decision Record)
- ✓ Use abstractions (like DI)
- ✓ Pin versions carefully
- ✓ Share usage patterns
- ✓ Setup dependency scans in CI/CD
- ✓ Design for replaceability
- ✓ Plan exit strategy

Operation Stage Checklist

- ✓ Patch management
- ✓ Monitor vulnerabilities
- ✓ Watch for project health
- ✓ Test boundaries
- ✓ Keep track of transitive deps

Transition Stage Checklist

- ☒ Handle vendor pivots
- ☒ Migrate behind abstractions
- ☒ Case study ready

Summing Up

1. 📁 The basics: Healthy project, , Maintained, Secure, Licensed, Documented
2. 🔍 4 deeper rules: Small, Isolated, Uncomplicated, Replaceable
3. 📁 Lifecycle checklists: Evaluate before adoption, Monitor and document, Plan for exit



Thanks! Questions?

Jonathan "J." Tower

🏆 Microsoft MVP in .NET

✉️ jtower@trailheadtechnology.com

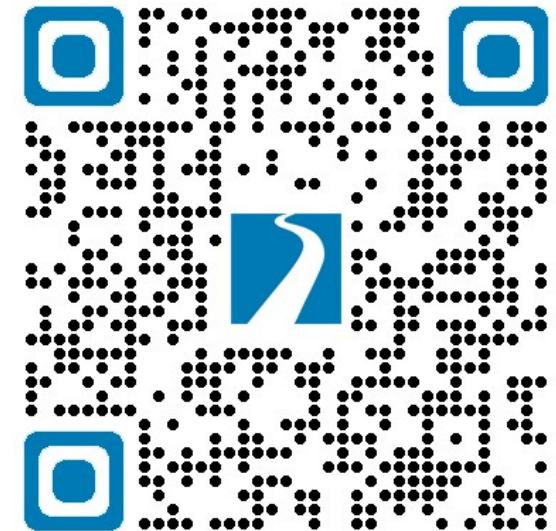
🌐 trailheadtechnology.com/blog

🐦 [jtowermi](https://twitter.com/jtowermi)

📘 [jtower](https://www.linkedin.com/in/jtower)

github.com/trailheadtechnology/3rd-party-lib

EXPERT CONSULTATION



bit.ly/th-offer